Czech Technical University
Faculty of Nuclear Sciences and Physical Engineering

Department of Mathematics

Applied Mathematical and Stochastic Methods

Adaptive Portfolio Optimization

MASTER'S THESIS

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Co-Supervisor: doc. Ing. Quang Van Tran Ph.D.
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ZADÁNÍ DIPLOMOVÉ PRÁCE

I. OSOBNÍ A STUDIJNÍ ÚDAJE

Příjmení: **Procházka** Jméno: **Tomáš** Osobní číslo: **509427**

Fakulta/ústav: **Fakulta jaderná a fyzikálně inženýrská**

Zadávající katedra/ústav: **Katedra matematiky**

Studijní program: **Aplikované matematicko-stochastické metody**

II. ÚDAJE K DIPLOMOVÉ PRÁCI

Název diplomové práce:

Adaptivní optimalizace portfolia

Název diplomové práce anglicky:

Adaptive Portfolio Optimization

Jméno a pracoviště vedoucí(ho) diplomové práce:

Ing. Miroslav Kárný, DrSc. ÚTIA AV ČR Praha

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III. PŘEVZETÍ ZADÁNÍ

Diplomant bere na vědomí, že je povinen vypracovat diplomovou práci samostatně, bez cizí pomoci, s výjimkou poskytnutých konzultací. Seznam použité literatury, jiných pramenů a jmen konzultantů je třeba uvést v diplomové práci.

Datum převzetí zadání

Bc. Procházka Tomáš

Podpis studenta

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Pokyny pro vypracování:

Obecně doplňte výzkum, který vedl k pracem [8,9], a rozsáhle ověřte dosaženou kvalitu rozhodování:
1. Upřesněte popis řešeného technického problému optimalizace portfolia [1,2,3].
2. Zlepšete algoritmus odhadu struktury pomocí hierarchického bayesovského odhadování užitého na sdružené apriorní rozdělení [4].
3. Formulujte a řešte optimalizační problém jako plně pravděpodobnostní návrh (PPN) rozhodovacích strategií [5,6]. Využijte vztah PPN ke kvadratické optimalizaci s váhami závislými na míře neurčitosti.
4. Vylepšete použité sekvenční kvadratické programování [4] s omezeními tak, aby nepotřebovalo heuristiku z [9] a zároveň zachovalo užití numericky stabilních Choleského odmocnin.
5. Implementujte získané adaptivní řešení založené na PPN a porovnejte ho s jeho deterministickým protějškem.
6. Experimentálně ověřte kvalitu celkového zpracování. Bude-li třeba použijte další techniky zvládající těžké chvosty v rozdělení dat [10] a nepřesnosti odhadů (opatrná strategie [11]).

Seznam doporučené literatury:

[1] A. Gunjan,·S. Bhattacharyya, A brief review of portfolio optimization techniques, Artificial Intelligence Review, 56:3847–3886, 2023.
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[8] T. Procházka, Probabilistic modelling in adaptive portfolio optimization. Bachelor thesis, FNSPE, CTU, Prague, 2024.
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[10] N. Khatun, et al. Applications of normality test in statistical analysis. Open journal of statistics, 11.01: 113, 2021.
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DECLARATION

I, the undersigned

Student's surname, given name(s):	Procházka Tomáš
Personal number:	509427
Programme name:	Applied Mathematical Stochastic Methods

hereby declare that the Master's Thesis titled

Adaptive Portfolio Optimization

is my own independent work and that I have declared all sources of information used in accordance with the Methodological Guideline for adhering to ethical principles when elaborating an academic final thesis and the Framework Rules for the use of artificial intelligence at CTU for study and pedagogical purposes in Bachelor and continuing Masters studies.

I declare that I have used artificial intelligence tools during the preparation and writing of the final thesis.
I've verified the generated content. I confirm that I am aware that I am fully responsible for the content of the final thesis.

In Prague on 07.05.2026

Bc. Tomáš Procházka
....
student's signature

Poděkování:

Děkuji svému školiteli Ing. Miroslavu Kárnému za trpělivost, přátelský přístup a za podporu během psaní závěrecné práce. Také děkuji svému spoluvedoucímu doc. Quang Van Tranovi za přínosné konzultace. Tento projekt je podpořen Joint Research Laboratory TALISMAN, EU-COST ACTION CA24136, SGS25/167/OHK4/3T/14 a SGS26/170/OHK4/3T/14.

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Název práce:

Adaptivní optimalizace portfolia

Autor: Bc. Tomáš Procházka

Studijní program: Applied Mathematical and Stochastic Methods

Druh práce: Diplomová práce

Vedoucí práce: Ing. Miroslav Kárný, DrSc., UTIA a.v. ČR
Spoluvedoucí práce: doc. Ing. Quang Van Tran Ph.D., FJFI, ČVUT v Praze

Abstrakt: Tato práce se věnuje vývoji schématu bayesovského učení pro autoregresivní modely s exogenními proměnnými a navržený přístup optimálně volí sdružené apriorní rozdělení neznámých parametrů, odhaduje strukturu modelu a sleduje časové změny parametrů pomocí datově řízeného zapomínání. Na tomto základě je formulován návrh vícekrokové rozhodovací strategie s kvadratickým kritériem, řešený pomocí optimalizace v maticových odmocninách a realizovaný v rámci adaptivní strategie s klouzavým horizontem. Výsledná strategie využívá plně pravděpodobnostní návrh, který zajišťuje konzistentní a metodologické řízení rizika a umožňuje návrh založený na Thompsonově vzorkování z distribuce parametrů. Tento postup poskytuje teoreticky kompletní analytické řešení náročného rozhodovacího problému a je aplikovatelný na širokou třídu příbuzných úloh. Počet hyperparametrů vyžadujících expertní ladění je malý. Experimenty na reálných datech demonstrují, že navržená strategie pro adaptivní optimalizaci portfolia překonává referenční rovnoměrné portfolio, což potvrzuje efektivitu celého přístupu.

Klíčová slova: optimalizace portfolia, pravděpodobnostní modelování, zapomínání, adaptivní rozhodování, plně pravděpodobnostní návrh

Title:

Adaptive Portfolio Optimization

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Abstract: This thesis develops a Bayesian learning framework for autoregressive models with exogenous variables, and the proposed approach optimally selects a conjugate prior for unknown parameters, infers the model structure, and tracks parameter variations through a data-driven optimized forgetting algorithm. Building on this model, the thesis introduces an expected multi-step decision policy design under quadratic loss, formulated via feasible square-root optimization and implemented in a receding-horizon setting. The policy exploits a fully probabilistic design that provides consistent and principled risk handling, enabling an off-line design strategy based on Thompson sampling of parameters. The resulting framework offers a theoretically complete and analytical solution to a challenging decision-making problem and is applicable to a broad class of related problems. The number of hyperparameters requiring expert tuning is low. Experiments on real-world data demonstrate that the proposed portfolio optimization strategy consistently outperforms the uniform portfolio benchmark under relevant and reasonable conditions, validating the effectiveness of the approach.

Key words: portfolio optimization, probabilistic modeling, forgetting, adaptive decision-making, fully probabilistic design

Contents

Notations	
$:=$	symbol for assigning value
$\propto$	proportionality symbol
$\mathbb{N}$	natural numbers
$\mathbb{R}$	real numbers
$\mathbb{R}^N$	the Cartesian product $\times_{i=1}^{N}\mathbb{R}$
$\mathbf{A} \in \mathbb{R}^{n,m}$	real (n, m)-matrix (rows, columns), $n, m \in \mathbb{N}$
$\mathbf{A}^\top$	transposition
$\lvert\mathbf{A}\rvert$	determinant
$\mathbf{O}$	zero matrix
$\mathbf{I}$	identity matrix
$\mathbf{1}$	vector of units
$\mathbf{tr}(\mathbf{A})$	trace
$\mathbf{A} > 0$	positive definite matrix
$\mathbf{A} \geq 0$	positive semi-definite matrix

Table 1: Notions and notations used throughout the document.

1 Introduction

This thesis presents a structured, methodological approach to the portfolio optimization problem. It is a continuation of author's previous work [30], which serves as a foundation for this thesis. It complements, improves upon and puts together the necessary parts needed for an adaptive, feasible and complete solution. A systematic and precise step-by-step description of the whole process is outlined in Chapter 5. For an easy reading, a pipeline diagram that captures the approach described in the text, is in Fig. 1.1. An entity that uses the scheme and receives the returns is referred to as user or investor.

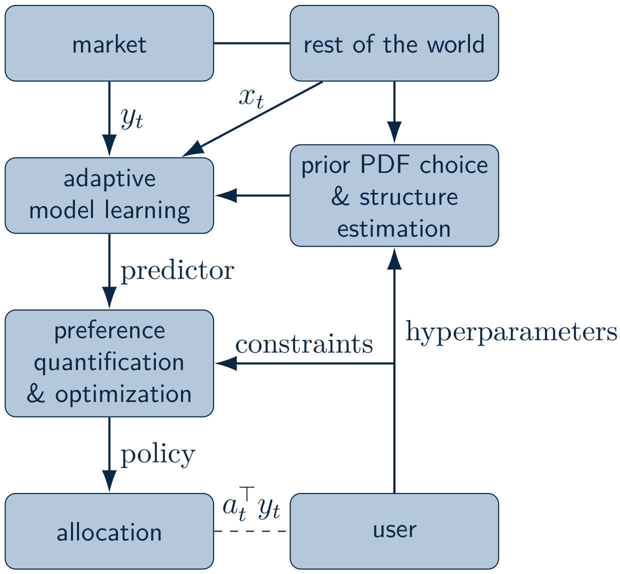


Figure 1.1: *Solution pipeline*: The market produces the observed y_t, which are returns of individual assets in the user's portfolio at time t. The rest of the world, which also concerns past market data, gives information x_t, potentially useful for predicting the said returns. Data serves for model structure estimation, choice of Bayesian prior and forgetting. The learned model (available knowledge) $\to y_t$ serves for a quantitative expression of the user's desires and for the optimal allocation a_t of the funds before y_t is realized. After observing y_t, the user receives the return $a_t^\top y_t$ minus the transaction costs.

We utilize many well-known practices and results from probability theory [4], Bayesian statistics [28], [9], quadratic optimization [12], linear-quadratic regulation (LQR) [29], especially in the fully probabilistic design (FPD) setting [17] and heuristics [3]. The main contributions of this thesis consist of unifying, improving, and connecting these areas to create an algorithmic approach to decision making that leverages a strong theoretical foundation with minimal need for user input.

The developed design pipeline[1] is useful not only for application on the financial market but also, with slight modifications, a wide variety of decision making tasks.

Research in the field of quantitative portfolio optimization spans a vast area of different views on the problem and solution methods. The most popular ones include classical approaches, like [26], that frame the problem as dual optimization, where expected returns are maximized and variance is minimized.

Innovative solutions, as in [1], for example, optimize the objective function, which is altered to meet user preferences like adversity to risk and diversification priorities directly, instead of fitting a market model. However, its complexity and number of possible configurations of the model that are left to be decided by the user hinder the model's decisive power.

The systematic way to the specific version of the preference elicitation problems, [10], is a significant part of our solution. It optimizes the expected quadratic loss expressing portfolio objectives. Together with the adopted, online estimated, autoregressive returns model with exogenous variables (ARX), [28], the solution becomes a version of the linear quadratic regulator task, which is common in the control domain.

A similar approach to the one taken in this thesis is used in article [24], where the authors use the LQR framework with linear predictor, updated via Kalman filter to achieve optimal control of a system with vector input and output. The cited article, however, concerns itself with controlling an input into an electrical motor, showing the broad applicability of such methods. One proposed enhancement of the LQR, [24] utilizes particle swarm optimization applied in the article [11] to optimize the efficient frontier, meaning the relationship between portfolio returns and their variance.

LQR setup also applies to related tasks to portfolio optimization. For example, high-frequency trading[2] is addressed as the continuous-time for hedging portfolios on the options[3] market. Unlike the multitude of academic papers, the solution in [29] considers transaction costs. For further reading about developed methods for portfolio optimization, including machine learning, see, for example, the review [13].

The range of applied methods and variability in results indicates that the problem is far from solved. This work aims to develop a solution method that does not leave aside any of the important sub-tasks, relies very little on user input, is as deductive as possible.

To properly formulate the problem, key variables and notation need to be defined. The returns of individual assets are observed in discrete time steps $t \in \mathbb{N}_0$, $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$ and are

[1] Python code with implemented pipeline and all of the presented algorithms is available at GitHub, https://github.com/ProchTomas/ResearchProject.

[2] 'High frequency' traders utilize algorithmic solutions to make transactions on the scale of milliseconds.

[3] Options are a very complicated financial tool and are out of the scope of this thesis.

denoted $y_{t,i}$, $y_t \in [-1, +\infty)^N$ being N-vector containing the returns of all N assets $(y_{t,i})_{i=1}^N$ for time t. The number of assets $N \in \mathbb{N}$, considered in our portfolio, at the time t of the allocation decision a_t, is taken as fixed and finite. This assumption is adopted without loss of generality. If the investor wishes to add a new asset to their portfolio, its previous weight in the portfolio is taken as zero. On the other hand, if they wish to delete an asset, the current weighing of the portfolio is recalculated before the optimization step.

All of the considered assets are denominated in United States Dollar, and the portfolio is optimized to achieve the highest percentage return. Currency risks[4]are not considered.

We use $x_t \in \mathbb{R}^\rho, \rho \geq N$, a ρ-vector of other observed variables, possibly also previous returns, at time t. $c_{t,i} \in [0, 1)$ are transaction costs associated with asset $i \in \{1, \dots, N\}$ at time t, $\mathbf{D}_t := \operatorname{diag}(c_{t,1}, \dots, c_{t,N})$. The concrete allocation of funds is held in the N-vector a_t. We are concerned only with a spot and long-only[5] portfolio. This poses constraints on the action space $\mathcal{A} = \{a \in \mathbb{R}^N \mid \mathbf{1}^\top a = 1, a \geq 0\}$. The collection of all available data up to and including the time step t, is denoted $\mathcal{D}^{(t)} = \mathcal{D}^{(t-1)} \cup \{y_t, x_t, a_t, c_t\}$, $\mathcal{D}^{(0)}$ formally marks prior knowledge.

The market returns are modeled by a multivariate linear regression model, Section 2.1. Its parameters are unknown and almost time-variant. The estimates of said parameters are updated in real time. We cover the choice of the prior probability density function (pdf), Section 2.4, the models' structure estimation, Section 2.5 and parameter tracking via adaptive, data-driven forgetting, Section 2.3. The gained linear model is subsequently used in the portfolio optimization, Chapter 4. The loss function in the optimization task quantitatively reflects the investor's wishes to maximize expected returns, while including the transaction costs and respecting uncertainty about the future returns. This specification exploits the FPD framework in decision strategies [18], Chapter 3, which fits the adopted Bayesian estimation [28] and helps to integrate uncertainty-induced risks.

The developed pipeline is tested on historical data, within reasonable assumptions and subsequently on real-time market data in Chapter 5. Conclusions of our work along with future trajectories worth exploring are discussed in Chapter 6.

[4]Assets denominated in different currencies other than the USD are universally recognized to carry higher risk of currency devaluation and/or inflation and/or the risk of higher volatility of the exchange rate against the USD.

[5]This means short-selling and leverage trading are prohibited.

2 Model for Returns

2.1 Multivariate Linear Regression

A multivariate, autoregressive linear model with exogenous variables is used for forecasting market returns y_t. This is a quite simple and well-studied, yet powerful model. It allows for an efficient Bayesian parameter estimation, and gives the returns' predictor. The ARX model reads

$$f_t = \mathbf{P}^\top x_t + e_t, \quad e_t \sim \mathcal{N}(0, \mathbf{\Sigma}), \tag{2.1}$$

where $\mathbf{P}$ is (ρ, N)-matrix of regression coefficients, x_t is a column ρ-vector of regressors. The N-vector e_t is normally distributed noise with zero mean and covariance $\mathbf{\Sigma}$, independent of x_t and e_τ for $\tau < t$. The column N-vector f_t is a function of returns y_t, possibly of actions a_t and previous data $\mathcal{D}^{(t-1)}$.

$$f_t = f_t(y_t, a_t, \mathcal{D}^{(t-1)}). \tag{2.2}$$

a_t is a real, column N-vector, that holds the allocation at time t.

Regular transformations between y_t and f_t are entertained only, with the non-zero Jacobian

$$J_f(\mathcal{D}^{(t)}) = \begin{vmatrix} \frac{\partial f_{t,1}}{\partial y_{t,1}} & \cdots & \frac{\partial f_{t,N}}{\partial y_{t,1}} \\ \vdots & & \vdots \\ \frac{\partial f_{t,1}}{\partial y_{t,N}} & \cdots & \frac{\partial f_{t,N}}{\partial y_{t,1}} \end{vmatrix}. \tag{2.3}$$

Matrices $\mathbf{P}$ and $\mathbf{\Sigma}$ form an unknown parameter $\theta := \{\mathbf{P}, \mathbf{\Sigma}\} \in \Theta$ of the model. The corresponding parametric model, pdf denoted through $p(.)$ for the joint, prior, posterior and predictive pdfs, $m(.)$ for the returns model and $r(.)$ for the decision rules, reads

$$m(y_t \mid a_t, \mathcal{D}^{(t-1)}, \theta) = (2\pi)^{-\frac{N}{2}} |\mathbf{\Sigma}|^{-\frac{1}{2}} \exp\left[-\frac{1}{2}(f_t - \mathbf{P}^\top x_t)^\top \mathbf{\Sigma}^{-1}(f_t - \mathbf{P}^\top x_t)\right] J_f(\mathcal{D}^{(t)}). \tag{2.4}$$

To obtain the predictor $p(y_t|a_t, \mathcal{D}^{(t-1)})$, that is needed for designing implementable decision rules for allocation a_t, we have to recall some basic relations from Bayesian parameter estimation [9].

The fundamental relation is the Bayes theorem:

$$p(\theta \mid \mathcal{D}^{(t)}) = \frac{p(\mathcal{D}^{(t)} \mid \theta) p(\theta)}{p(\mathcal{D}^{(t)})}. \tag{2.5}$$

Applying marginalization, it reads

$$p(\theta \mid \mathcal{D}^{(t)}) = \frac{p(\mathcal{D}^{(t)} \mid \theta)p(\theta)}{\int_\Theta p(\mathcal{D}^{(t)} \mid \theta)p(\theta)d\theta}, \tag{2.6}$$

the prior knowledge $\mathcal{D}^{(0)}$ is implicitly present in all conditions. To express (2.6), we use the chain rule

$$p(\mathcal{D}^{(t)} \mid \theta) = \prod_{\tau=1}^{t} m(y_\tau \mid a_\tau, \mathcal{D}^{(\tau-1)}, \theta) r(a_\tau \mid \mathcal{D}^{(\tau-1)}, \theta) \tag{2.7}$$

and the natural conditions of control [28], $(r(a_\tau|\mathcal{D}^{(\tau-1)}, \theta) = r(a_\tau \mid \mathcal{D}^{(\tau-1)}))_{\tau=1}^{h}$, where h is the considered decision horizon $h \in \mathbb{N}$. This means that θ is unknown to the action generator. Then (2.6) becomes

$$p(\theta \mid \mathcal{D}^{(t)}) = \frac{l(\mathcal{D}^{(t)}, \theta)p(\theta)}{\int_\Theta l(\mathcal{D}^{(t)}, \theta)p(\theta)d\theta},$$

where

$$l(\mathcal{D}^{(t)}, \theta) = \prod_{\tau=1}^{t} m(y_\tau \mid a_\tau, \mathcal{D}^{(\tau-1)}, \theta) \tag{2.8}$$

is the likelihood function. Using the definition of the likelihood function and applying marginalization again, we arrive at the predictor of the future returns

$$m(y_{t+1} \mid a_{t+1}, \mathcal{D}^{(t)}) = \frac{\int_\Theta l(\mathcal{D}^{(t+1)}, \theta)p(\theta)d\theta}{\int_\Theta l(\mathcal{D}^{(t)}, \theta)p(\theta)d\theta}. \tag{2.9}$$

The likelihood (2.8), with the pdf (2.4) can be expressed as

$$l(\mathcal{D}^{(t)}, \theta) = (2\pi)^{-N\frac{t}{2}} |\mathbf{\Sigma}|^{-\frac{t}{2}} \exp\left[-\frac{1}{2}\mathrm{tr}\left(\mathbf{\Sigma}^{-1}\begin{pmatrix}-\mathbf{I} & \mathbf{P}^\top\end{pmatrix}\tilde{\mathbf{V}}_t\begin{pmatrix}-\mathbf{I}\\ \mathbf{P}\end{pmatrix}\right)\right]\prod_{\tau=0}^{t} J_f(\mathcal{D}^{(\tau)}). \tag{2.10}$$

The extended information matrix $\tilde{\mathbf{V}}_t$ and the counting statistics t form a sufficient statistics. This sufficient statistics encapsulates the collected data and can be combined with the functional form reproducing prior pdf $p(\theta) = p(\theta|\mathbf{V}_0, \Delta_0)$, $\mathcal{D}^0 = (\mathbf{V}_0, \Delta_0)$,

$$\mathbf{V}_t = \sum_{\tau=0}^{t}\begin{pmatrix}f_\tau\\ x_\tau\end{pmatrix}\begin{pmatrix}f_\tau^\top & x_\tau^\top\end{pmatrix} + \mathbf{V}_0 = \tilde{\mathbf{V}}_t + \mathbf{V}_0, \quad \Delta_t := t + \Delta_0. \tag{2.11}$$

The matrix $\mathbf{V}_0 > 0, \Delta_0 > \rho$, expresses the initial belief $p(\theta)$ about θ quantified by the conjugate Gauss-inverse-Wishart pdf. The optimized prior choice of $\mathbf{V}_0, \Delta_0$ is discussed in Section 2.4. The matrix (2.11) is partitioned as

$$\mathbf{V}_t = \begin{pmatrix}\mathbf{V}_{f,t} & \mathbf{V}_{xf,t}^\top\\ \mathbf{V}_{xf,t} & \mathbf{V}_{x,t}\end{pmatrix} \tag{2.12}$$

and the prior $\mathbf{V}_0$ is partitioned the same way. It allows us to relate this statistics to the least squares estimate $\hat{\mathbf{P}}_t$ of the matrix of regression coefficients. If $\mathbf{V}_0 = \mathbf{O}$, $\hat{\mathbf{P}}_t$ solves

$$\min_{\mathbf{P}} ||(f_0, f_1, \ldots, f_t) - \mathbf{P}^\top(x_0, x_1, \ldots, x_t)||^2,$$

and has the familiar form

$$\hat{\mathbf{P}}_t^\top = \left((x_0, x_1, \ldots, x_t)(x_0, x_1, \ldots, x_t)^\top\right)^{-1} (x_0, x_1, \ldots, x_t)(f_0, f_1, \ldots f_t)^\top = \mathbf{V}_{x,t}^{-1}\mathbf{V}_{xf,t}. \tag{2.13}$$

The least squares remainder reads

$$\mathbf{\Lambda}_t = \mathbf{V}_{f,t} - \mathbf{V}_{xf,t}^\top \mathbf{V}_{x,t}^{-1}\mathbf{V}_{xf,t}. \tag{2.14}$$

When the block form (2.12), the estimate of the matrix of regression coefficients (2.13) and the remainder (2.14), using these blocks, are applied to the matrix $\mathbf{V}_0 > 0$, the a posteriori likelihood with the considered prior can be rewritten into

$$l(\mathcal{D}^{(t)}, \theta)p(\theta) = \frac{\prod_{\tau=1}^t J_f(\mathcal{D}^{(t)})}{(2\pi)^{N\frac{\Delta_t}{2}}|\mathbf{\Sigma}|^{\frac{\Delta_t}{2}}} \exp\left[-\frac{1}{2}\mathrm{tr}\left(\mathbf{\Sigma}^{-1}(\mathbf{P} - \hat{\mathbf{P}}_t)^\top \mathbf{V}_{x,t}(\mathbf{P} - \hat{\mathbf{P}}_t) + \mathbf{\Sigma}^{-1}\mathbf{\Lambda}_t\right)\right]. \tag{2.15}$$

The posterior pdfs of the unknown parameters are

$$p(\mathbf{P} \mid \mathcal{D}^{(t)}, \mathbf{\Sigma}) = (2\pi)^{-\frac{\rho N}{2}}|\mathbf{V}_{x,t}|^{\frac{N}{2}}|\mathbf{\Sigma}|^{-\frac{\rho}{2}} \exp\left[-\frac{1}{2}\mathrm{tr}\left(\mathbf{\Sigma}^{-1}(\mathbf{P} - \hat{\mathbf{P}}_t)^\top \mathbf{V}_{x,t}(\mathbf{P} - \hat{\mathbf{P}}_t)\right)\right], \tag{2.16}$$

$$p(\mathbf{\Sigma}^{-1} \mid \mathcal{D}^{(t)}) = \frac{|\mathbf{\Lambda}_t|^{\frac{\Delta_t - \rho + N + 1}{2}}|\mathbf{\Sigma}|^{-\frac{\Delta_t - \rho}{2}}}{(2^{\Delta_t - \rho + N + 1}\pi^{\frac{N-1}{2}})^{\frac{N}{2}} \prod_{j=1}^N \Gamma\left(\frac{\Delta_t - \rho + N + 2 - j}{2}\right)} \exp\left[-\frac{1}{2}\mathrm{tr}\left(\mathbf{\Sigma}^{-1}\mathbf{\Lambda}_t\right)\right]. \tag{2.17}$$

These distributions will be useful further on. The maxima of posterior distributions 2.16, 2.17 are achieved at $\mathbf{P} = \hat{\mathbf{P}}_t$ and $\hat{\mathbf{\Sigma}}_t^{-1} = \Delta_t \mathbf{\Lambda}_t^{-1}$, (2.13), (2.14) respectively.

It is often the case that the sub-matrix $\mathbf{V}_x$ is poorly conditioned due to near collinearity in regressors x_t. This is resolved by keeping and updating the matrix $\mathbf{V}_t$ as its positive semi-definite square root. An efficient algorithm for such updating, based on the Cholesky decomposition is given in [27]. It takes

$$\mathbf{V}_t^{-1} := \mathbf{L}_t \mathbf{L}_t^\top, \tag{2.18}$$

where $\mathbf{L}$ is a lower-triangular matrix. Other possible decompositions are discussed in [7], for example, the commonly used $LDL^\top$ decomposition is in previous work, [30] (2.18). The algorithm REFIL, presented in [27], has been used so far for updating the decomposition. However efficient, this algorithm may introduce considerable numerical errors in the calculation. Moreover, the inversion in (2.18) is needed asynchronously with data updating.

The decomposition fitting our problem handling is

$$\mathbf{V}_t = \mathbf{G}_t \mathbf{G}_t^\top, \tag{2.19}$$

where $\mathbf{G}_t$ is an upper triangular matrix. The block form of the square root is partitioned as

$$\mathbf{G}_t \mathbf{G}_t^\top = \begin{pmatrix} \mathbf{G}_{f,t} & \mathbf{G}_{xf,t} \\ \mathbf{O} & \mathbf{G}_{x,t} \end{pmatrix} \begin{pmatrix} \mathbf{G}_{f,t}^\top & \mathbf{O} \\ \mathbf{G}_{xf,t}^\top & \mathbf{G}_{x,t}^\top \end{pmatrix} = \begin{pmatrix} \mathbf{G}_{f,t}\mathbf{G}_{f,t}^\top + \mathbf{G}_{xf,t}\mathbf{G}_{xf,t}^\top & \mathbf{G}_{xf,t}\mathbf{G}_{x,t}^\top \\ \mathbf{G}_{x,t}\mathbf{G}_{xf,t}^\top & \mathbf{G}_{x,t}\mathbf{G}_{x,t}^\top \end{pmatrix},$$

from which we derive the relations for the point estimate of regression coefficients (2.13) and covariance (2.14), maximizing a posteriori likelihood. The maximum of (2.15) is reached by minimizing over $\mathbf{P}$

$$\left\| \begin{pmatrix} \mathbf{G}_{f,t}^\top & \mathbf{O} \\ \mathbf{G}_{xf,t}^\top & \mathbf{G}_{x,t}^\top \end{pmatrix} \begin{pmatrix} -\mathbf{I} \\ \mathbf{P} \end{pmatrix} \right\|^2.$$

It is solved by

$$\hat{\mathbf{P}}_t = \mathbf{G}_{x,t}^{-\top}\mathbf{G}_{xf,t}^{\top}, \tag{2.20}$$

giving the remainder

$$\mathbf{\Lambda}_t = \mathbf{G}_{f,t}\mathbf{G}_{f,t}^{\top}, \tag{2.21}$$

and a point estimate of $\mathbf{\Sigma}$, $\hat{\mathbf{\Sigma}}_t = \frac{\mathbf{\Lambda}_t}{\Delta_t}$. The update only involves one orthogonal rotation, and its numerical stability is higher compared to REFIL. This simpler decomposition is also needed in the new update rule, realizing (2.11), that is presented in Section 2.3.

We also use Lemma *A.2* to update the determinants of $\mathbf{V}_x$ and $\mathbf{\Lambda}$. The update rule (2.11) then gives

$$|\mathbf{V}_{x,t}| = |\mathbf{V}_{x,t-1}|(1 + x_t^{\top}\mathbf{V}_{x,t-1}^{-1}x_t) = |\mathbf{G}_{x,t-1}|^2(1 + x_t^{\top}\mathbf{G}_{x,t-1}^{-\top}\mathbf{G}_{x,t-1}^{-1}x_t), \tag{2.22}$$

$$|\mathbf{\Lambda}_t| = |\mathbf{\Lambda}_{t-1}|\left(1 + \frac{\hat{e}_t^{\top}\mathbf{\Lambda}_{t-1}^{-1}\hat{e}_t}{1 + x_t^{\top}\mathbf{V}_{x,t-1}^{-1}x_t}\right) = |\mathbf{G}_{f,t-1}|^2\left(1 + \frac{\hat{e}_t^{\top}\mathbf{G}_{f,t-1}^{-\top}\mathbf{G}_{f,t-1}^{-1}\hat{e}_t}{1 + x_t^{\top}\mathbf{G}_{x,t-1}^{-\top}\mathbf{G}_{x,t-1}^{-1}x_t}\right), \tag{2.23}$$

where $\hat{e}_t = f_t - \hat{\mathbf{P}}_{t-1}^{\top}x_t$ is the prediction error.

2.2 Transformations

In this Section, we briefly discuss possible transformations (2.2). Their possible use is motivated by the fact that the distribution of market returns is known to have heavier tails than the assumed normal distribution, while also being confined to the interval $(-1, +\infty)$. This has made us inspect possibilities to model the transformed returns. A logarithmic transformation with a shift may be useful to respect the above facts

$$f_{t,i} = \ln(1 + y_{t,i}). \tag{2.24}$$

If f is modeled by a normally distributed random variable then $e^f = y + 1$ has a log-normal distribution. The expectation and covariance of y are found using the moment generating function, which gives

$$\mathrm{E}[y_{t,i}|\mathcal{D}^{(t-1)}] = \exp\left[\hat{f}_i + \frac{1}{2}\hat{\mathbf{\Sigma}}_{f,ii}\right] - 1$$

$$\hat{\mathbf{\Sigma}}_{y,ij} = \exp\left[\hat{f}_i + \hat{f}_j + \frac{1}{2}\left(\hat{\mathbf{\Sigma}}_{f,ii} + \hat{\mathbf{\Sigma}}_{f,jj}\right)\right]\left(\exp\left[\hat{\mathbf{\Sigma}}_{f,ij}\right] - 1\right),$$

where $\hat{\mathbf{\Sigma}}$ is the covariance estimate, given the transformed observations $f = \ln(1 + y)$, with point estimate $\hat{f}$.

The moments play a key role in the analytical solution to the optimization problem, Chapter 4. Taking a linear approximation of those moments, which is needed in the derived analytical solution for the optimal quadratic control, introduces an additional and unnecessary error. This is not appreciated and makes us abandon this possibility.

Another possibility is to use Generalized Linear Model (GLM) with the Gamma distribution and identity link function, applied on shifted returns $y_{t,i}+1$. The Gamma distribution addresses the boundedness to $(0, +\infty)$, skewed tails and heteroskedasticity, and it gives a linear expectation $\mathrm{E}[y_t|\mathcal{D}^{(t-1)}] = -1 + \hat{\mathbf{P}}_{t-1}^{\top}x_t$. However, Gamma GLM discards the Gaussian noise for the response and assumes the response follows a Gamma distribution instead. The simple sufficient statistics matrix (2.11) would no longer be viable.

Bayesian GLMs typically require numerical methods like Markov Chain Monte Carlo (MCMC), [14] or Variational Inference for posterior estimation, which are computationally intensive and do not fit the recursive, analytical structure we aim to build.

All this adds to the need to get analytically feasible input into dynamic programming. Therefore, we believe, a Gaussian model without any transformations of the observed market returns, is a meaningful choice, given the requirements.

While we know that the Gaussian model of returns does not capture the bounded nature, nor does it capture the heavy tails of the empirical distribution, this simplification is necessary to utilize the well-established framework of Bayesian linear regression and derive a closed-form LQR-style solution.

The acceptability of our choice $f_t = y_t$ is evaluated by the empirical performance of the resulting strategy.

2.3 Forgetting

The multivariate ARX model in Section 2.1 assumes constant parameters $\mathbf{P}$ and $\mathbf{\Sigma}$. This assumption generally does not hold for dynamic systems such as financial markets.

Forgetting irrelevant knowledge provides a simple, flexible way of estimating slowly varying parameters, which is quite helpful when dealing with financial markets, that are understood to have non-constant variance. The chosen forgetting structure modifies the update of the sufficient statistic (2.11) as follows

$$\mathbf{G}_t\mathbf{G}_t^\top = (\alpha+\beta)\mathbf{G}_{t-1}\mathbf{G}_{t-1}^\top + \beta\begin{pmatrix} f_t \\ x_t \end{pmatrix}\begin{pmatrix} f_t^\top & x_t^\top \end{pmatrix} + (1-\alpha-\beta)\mathbf{G}_0\mathbf{G}_0^\top, \tag{2.25}$$

$$\Delta_t = (\alpha+\beta)\Delta_{t-1} + \beta + (1-\alpha-\beta)\Delta_0, \quad \alpha,\beta \geq 0, \alpha+\beta \leq 1$$

The justification for such a choice of forgetting parameters will be explained here, together with data-driven choice for the forgetting parameters α, β, γ.

Method for adaptive estimation of these forgetting factors based on the minimum relative entropy principle is presented in [20] and applied to Gaussian distributions. Having a collection of possible relevant statistics $((\mathbf{V}_1, \Delta_1), (\mathbf{V}_2, \Delta_2), \dots)$ that determine the posterior Gauss-inverse-Wishart pdfs, the optimal values of their weights $\varphi = (\varphi_1, \varphi_2, \dots)$ in their weighted geometric mean are obtained by minimizing

$$F(\varphi) = D(\varphi||\varphi_0) - \ln\left[\int_\Theta \prod_i \left(p(\theta \mid \mathbf{V}_i, \Delta_i)\right)^{\varphi_i} d\mathbf{P}d\mathbf{\Sigma}\right], \tag{2.26}$$

where $D(\varphi||\varphi_0)$ is the Kullback-Leibler divergence (KLD) of probabilistic vectors φ, φ_0, [23] and

$$p(\theta \mid \mathbf{V}_i, \Delta_i) = p(\mathbf{P}, \mathbf{\Sigma}^{-1} \mid \mathbf{V}_i, \Delta_i) = p(\mathbf{P} \mid \mathbf{V}_i, \Delta_i, \mathbf{\Sigma})p(\mathbf{\Sigma}^{-1} \mid \mathbf{V}_i, \Delta_i)$$

from the equations (2.16), (2.17). $\mathbf{V}_i, \Delta_i$ are considered sufficient statistics corresponding to variants $\mathcal{D}^{(t_i)}$, from differing t_i. φ_0 is the prior guess for φ. We seek to find such

$$\varphi^* \in \arg\min_{\Omega_\varphi} F(\varphi), \quad \Omega_\varphi = \left\{ \varphi_i \geq 0, \sum_i \varphi_i = 1 \right\}.$$

We show how we obtained the update rule (2.25), combining $(\mathbf{V}_i, \Delta_i)_{i=1}^3$. We consider $t_1 = t, t_2 = t-1, t_3 = 0$, respectively, for the current possible change of the parameters. The choice t_1, t_2 represents the one-step-ahead update. The added term $t_3 = 0$ admits that the change is so large, that reversion to prior statistic may be worthwhile. The pdfs in function (2.26) are weighted by φ_i. The old information $\mathbf{V}_{t-1}$ is weighted by $\varphi_1 = \alpha$, and the newly observed information $\mathbf{V}_t$ by $\varphi_2 = \beta$. Now, a regularization term $\mathbf{V}_0$, from Section 2.4, is introduced, weighted by $\varphi_3 = \gamma$, $\gamma = 1 - \alpha - \beta$.

The resulting combination is $\alpha\mathbf{V}_{t-1} + \beta\mathbf{V}_t + \gamma\mathbf{V}_0$, which leads to (2.25). Recall the function (2.26), the joint pdf for the parameters, θ is

$$p(\mathbf{P}, \mathbf{\Sigma}^{-1} \mid \mathbf{V}_t, \Delta_t) = \frac{|\mathbf{V}_{x,t}|^{\frac{N}{2}} |\mathbf{\Sigma}^{-1}|^{\frac{\Delta_t}{2}} \exp\left[-\frac{1}{2}\mathrm{tr}\left(\mathbf{\Sigma}^{-1}(\mathbf{P} - \hat{\mathbf{P}})^\top \mathbf{V}_{x,t}(\mathbf{P} - \hat{\mathbf{P}}) + \mathbf{\Sigma}^{-1}\mathbf{\Lambda}\right)\right]}{(2\pi)^{\frac{N\rho}{2}} (2^{\Delta_t - \rho + N + 1}\pi^{\frac{N-1}{2}})^{\frac{N}{2}} \prod_{j=1}^N \Gamma(\frac{\Delta_t - \rho + N + 2 - j}{2})}, \tag{2.27}$$

multiplying $p(\theta \mid \mathbf{V}_{t-1}, \Delta_{t-1})^\alpha p(\theta \mid \mathbf{V}_t, \Delta_t)^\beta p(\theta \mid \mathbf{V}_0, \Delta_0)^\gamma$, the resulting term under the integral (2.26) is

$$\prod_{i=1}^3 p(\theta \mid \mathbf{V}_i, \Delta_i)^{\varphi_i} \propto |\mathbf{\Sigma}^{-1}|^{\frac{\tilde{\Delta}}{2}} \exp\left[-\frac{1}{2}\mathrm{tr}\left(\mathbf{\Sigma}^{-1}(\mathbf{P} - \hat{\mathbf{P}})^\top \tilde{\mathbf{V}}_x(\mathbf{P} - \hat{\mathbf{P}}) + \mathbf{\Sigma}^{-1}\tilde{\mathbf{\Lambda}}\right)\right],$$

where $\tilde{\mathbf{V}}_x = \alpha\mathbf{V}_{x,t-1} + \beta\mathbf{V}_{x,t} + \gamma\mathbf{V}_{x,0}$, $\tilde{\mathbf{\Lambda}} = \alpha\mathbf{\Lambda}_{t-1} + \beta\mathbf{\Lambda}_t + \gamma\mathbf{\Lambda}_0$ and $\tilde{\Delta} := \alpha\Delta_{t-1} + \beta\Delta_t + \gamma\Delta_0$. The integral is evaluated by first applying Lemma *A.4*

$$\int_\Theta \prod_{i=1}^3 p(\theta \mid \mathbf{V}_i, \Delta_i)^{\varphi_i} d\mathbf{P} d\mathbf{\Sigma} \propto (2\pi)^{\frac{N\rho}{2}} |\tilde{\mathbf{V}}_x|^{-\frac{N}{2}} \int_{M_+} |\mathbf{\Sigma}^{-1}|^{\frac{-\rho + \alpha\Delta_{t-1} + \beta\Delta_t + \gamma\Delta_0}{2}} \exp\left[-\frac{1}{2}\mathrm{tr}(\mathbf{\Sigma}^{-1}\tilde{\mathbf{\Lambda}})\right] d\mathbf{\Sigma}$$

and then Lemma *A.5*, which yields the result,

$$-\ln[\mathrm{J}(\tilde{\Delta}, \tilde{\mathbf{V}})] + \alpha\ln[\mathrm{J}(\Delta_{t-1}, \mathbf{V}_{t-1})] + \beta\ln[\mathrm{J}(\Delta_t, \mathbf{V}_t)] + \gamma\ln[\mathrm{J}(\Delta_0, \mathbf{V}_0)],$$

where

$$\mathrm{J}(\Delta, \mathbf{V}) = (2\pi)^{\frac{N\rho}{2}} |\mathbf{V}_x|^{-\frac{N}{2}} |\mathbf{\Lambda}|^{-\frac{\Delta - \rho + N + 1}{2}} 2^{\frac{N(\Delta - \rho + N + 1)}{2}} \prod_{j=1}^N \Gamma\left(\tfrac{\Delta - \rho + N + 2 - j}{2}\right)$$

is the normalizing constant of the pdf (2.27). Now everything is put together by adding the KLD with the probability mass functions φ_i. Then we omit any terms that do not depend on

α, β, γ. And the suitable form of (2.26) for calculations is gained

$$
\begin{aligned}
F(\alpha,\beta,\gamma) = {} & \alpha \ln\left[\frac{\alpha}{\alpha_0}\right] + \beta \ln\left[\frac{\beta}{\beta_0}\right] + \gamma \ln\left[\frac{\gamma}{\gamma_0}\right] \\
& \sum_{j=1}^{N} - \ln\Gamma\left(\tfrac{\tilde{\Delta}-\rho+N+2-j}{2}\right) + \alpha \ln\Gamma\left(\tfrac{\Delta_{t-1}-\rho+N+2-j}{2}\right) + \beta \ln\Gamma\left(\tfrac{\Delta_{t-1}+1-\rho+N+2-j}{2}\right) \\
& + \gamma \ln\Gamma\left(\tfrac{\Delta_0-\rho+N+2-j}{2}\right) \\
& - (\alpha+\beta)N \ln|\mathbf{G}_{x,t-1}| - \tfrac{\beta N}{2} \ln\left[1+\zeta_t\right] - \gamma N \ln|\mathbf{G}_{x,0}| + \tfrac{N}{2} \ln|\tilde{\mathbf{V}}_x| \\
& - \left[(\alpha+\beta)(\Delta_{t-1}-\rho+N+1)+\beta\right] \ln|\mathbf{G}_{f,t-1}| \\
& - \beta \tfrac{\Delta_{t-1}-\rho+N+1}{2} \ln\left[1+\tfrac{\kappa_t}{1+\zeta_t}\right] - \gamma(\Delta_0-\rho+N+1) \ln|\mathbf{G}_{f,0}| + \tfrac{\tilde{\Delta}-\rho+N+1}{2} \ln|\tilde{\mathbf{\Lambda}}|,
\end{aligned}
\tag{2.28}
$$

where $\zeta_t = x_t^\top \mathbf{G}_{x,t-1}^{-\top} \mathbf{G}_{x,t-1}^{-1} x_t$, $\kappa_t = \hat{e}_t^\top \mathbf{G}_{f,t-1}^{-\top} \mathbf{G}_{f,t-1}^{-1} \hat{e}_t$ from (2.22) and (2.23), respectively. In the implementation, we can use the decomposed matrices because $\tilde{\mathbf{V}}_x$ is a convex combination of positive definite matrices, and thus it is positive definite.

The parameters α, β that minimize the function (2.28), are found using the Nelder-Mead method, with projection onto a simplex to meet the constraints $\alpha, \beta, \gamma \geq 0$, $\alpha+\beta+\gamma = 1$. Nelder-Mead is an unconstrained optimization method, therefore, instead of optimizing on an unconstrained plain $z_1^{(i)}, z_2^{(i)}, z_3^{(i)} \in \mathbb{R}$, we first re-parametrize to $\exp(z_1), \exp(z_2), 1$, then

$$
\alpha^{(i)} = \frac{\exp\left[z_1^{(i)}\right]}{\exp\left[z_1^{(i)}\right] + \exp\left[z_2^{(i)}\right] + 1}, \beta^{(i)} = \frac{\exp\left[z_2^{(i)}\right]}{\exp\left[z_1^{(i)}\right] + \exp\left[z_2^{(i)}\right] + 1}, \gamma^{(i)} = \frac{1}{\exp\left[z_1^{(i)}\right] + \exp\left[z_2^{(i)}\right] + 1}
$$

in each iteration (i) of the method. Then, the sufficient statistics is updated according to the rule (2.25) written in non-square-root version

$$
\mathbf{V}_t := (\alpha+\beta)\mathbf{V}_{t-1} + \beta \begin{pmatrix} f_t \\ x_t \end{pmatrix} \begin{pmatrix} f_t^\top & x_t^\top \end{pmatrix} + (1-\alpha-\beta)\mathbf{V}_0. \tag{2.29}
$$

This formula has a neat interpretation. The old information should get higher or at least the same value as the new one. If the new information and the old one are no longer accurate, we revert to the prior $\mathbf{V}_0$, which can also be seen as a regularization term.

The update with simultaneous decomposition that gives (2.19) is done with orthogonal transformations using the RQ[6] algorithm

$$
\mathbf{G}_t \mathbf{G}_t^\top = (\alpha+\beta)\mathbf{G}_{t-1}\mathbf{G}_{t-1}^\top + \beta \begin{pmatrix} f_t \\ x_t \end{pmatrix} \begin{pmatrix} f_t^\top & x_t^\top \end{pmatrix} + (1-\alpha-\beta)\mathbf{G}_0\mathbf{G}_0^\top =
$$

$$
= \underbrace{\begin{pmatrix} \sqrt{\alpha+\beta}\,\mathbf{G}_{t-1} & \sqrt{\beta}\begin{pmatrix} f_t \\ x_t \end{pmatrix} & \sqrt{1-\alpha-\beta}\,\mathbf{G}_0 \end{pmatrix} \mathbf{R}}_{\mathbf{G}_t} \underbrace{\mathbf{R}^\top \begin{pmatrix} \sqrt{\alpha+\beta}\,\mathbf{G}_{t-1}^\top \\ \sqrt{\beta}\begin{pmatrix} f_t^\top & x_t^\top \end{pmatrix} \\ \sqrt{1-\alpha-\beta}\,\mathbf{G}_0^\top \end{pmatrix}}_{\mathbf{G}_t^\top},
$$

[6] RQ algorithm is equivalent to the QR algorithm.

$$\Delta_t = (\alpha+\beta)\Delta_{t-1} + \beta + \gamma\Delta_0,$$

where $\mathbf{R}\mathbf{R}^\top = \mathbf{I}$, $\mathbf{R}$ is a unitary rotation matrix.

Algorithm 1 Update Upper-Triangular Factor $\mathbf{G}$ and Δ

Require: $\mathbf{G}$, f, x, $\mathbf{G}_0$, α, β, Δ, Δ_0

$\Delta \leftarrow (\alpha+\beta)\Delta + \beta + (1-\alpha-\beta)\Delta_0$

if $\beta = 1$ **then** ▷ Case: no forgetting

$\quad \mathbf{M} \leftarrow \begin{pmatrix} \begin{pmatrix} f \\ x \end{pmatrix} & \mathbf{G} \end{pmatrix}$

else

$\quad \mathbf{M} \leftarrow \begin{pmatrix} \sqrt{\alpha+\beta}\mathbf{G} & \sqrt{\beta}\begin{pmatrix} f \\ x \end{pmatrix} & \sqrt{1-\alpha-\beta}\mathbf{G}_0 \end{pmatrix}$

end if

$\mathbf{R} \leftarrow \mathrm{rq}(\mathbf{M})$ ▷ rq is a "transposed" QR algorithm

$diag \leftarrow diag(R)$

$signs = \mathrm{where}(diag \geq 0)$

$\mathbf{G} \leftarrow \mathbf{R} * signs$ ▷ Scale columns of $\mathbf{R}$ to ensure positive diagonal

return $\mathbf{G}$, Δ

Example 2.1.

The potential of this approach is demonstrated on a simple parameter estimation problem, where $y_t = f_t$ is a scalar random variable and $x_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $x_t \in \mathbb{R}^3$, $\forall t$. The standard deviation of random noise e_t has two components, a constant and time-dependent sinus function, $e_t \sim \mathcal{N}(0, (0.1+\sin(4\pi t))^2)$. $y_t = \mathbf{P}x_t + (e_t + \mathrm{vol}_t)$, where a volatility component vol_t is introduced to more closely simulate the conditions on the markets. It is generated by randomly choosing small time intervals in the simulated data y, where we add or subtract a random number outside the expected data values. The volatility simulates black-swan events[7]. The adopted forgetting is compared to a standard REFIL, (2.18) where the update rule uses constant exponential forgetting

$$\mathbf{V}_t = (\alpha_0 + \beta_0)\mathbf{V}_{t-1} + \begin{pmatrix} f_t \\ x_t \end{pmatrix} \begin{pmatrix} f_t^\top & x_t^\top \end{pmatrix}.$$

The prior guesses of α, β, γ in (2.28) are set to $\alpha_0 = 0.039, \beta_0 = 0.96, \gamma_0 = 0.001$, reflecting our prior belief, that the new information is valuable. The relatively flat prior with $\mathbf{V}_0 = 0.1\mathbf{I}$ and $\Delta_0 = 4$ is chosen The simulation runs for 100 steps. Random seed 0 in numpy[8] was used to generate the data. The evolution of the parameter estimate $\hat{\mathbf{P}}_t$ are shown in Figs. 2.1, 2.2.

[7] A black-swan event is a fundamentally unpredictable large move. It usually has a negative connotation.
[8] All algorithms are coded in Python.

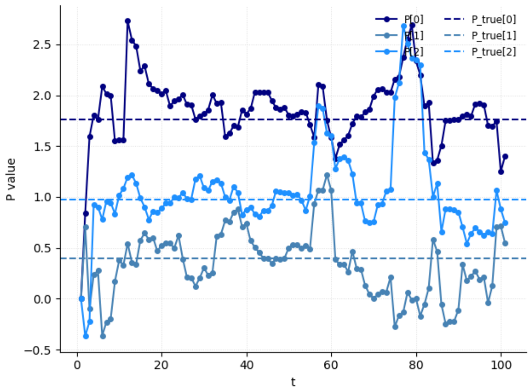


Figure 2.1: Evolution of point estimate $\hat{\mathbf{P}}_t$

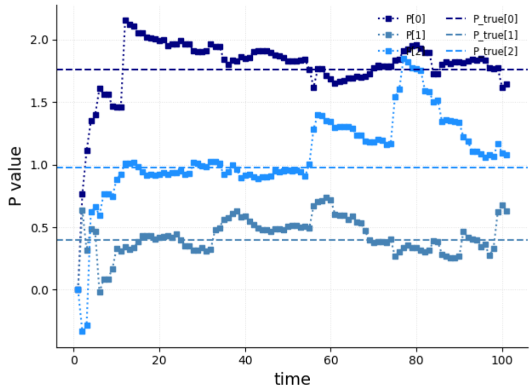


Figure 2.2: Evolution of point estimate $\hat{\mathbf{P}}_{t.\text{REFIL}}$

The inability to predict black-swan events is to be expected, the interesting part is that the adopted forgetting is able to pick up on such extremes and revert back to the norm fairly quickly, compared to standard REFIL. The metrics used to evaluate this model are the sign accuracy and mean square error

$$\text{sgn-acc} = \frac{1}{t}\sum_{\tau=1}^{t} \chi(\text{sgn}(\hat{y}_\tau) = \text{sgn}(y_\tau)), \quad \hat{y}_\tau = \hat{\mathbf{P}}_{\tau-1} x_\tau$$

$$\text{MSE} = \frac{1}{t}\sum_{\tau=1}^{t} (\hat{y}_\tau - y_\tau)^2$$

where the function $\chi(.)$ is a boolean indicator. Resulting MSE = 7.54, sgn-acc = 86%, in this example. REFIL produced the same sign accuracy, while its MSE = 7.66. REFIL captures the trend and provides arguably more stable estimates of the parameter values, however, as seen in Fig. 2.4 it fails to capture the simulated behavior. These properties make REFIL a good candidate for systems well-modeled with the ARX having slowly varying parameters. The implemented forgetting has the potential to exceed it in more volatile environments.

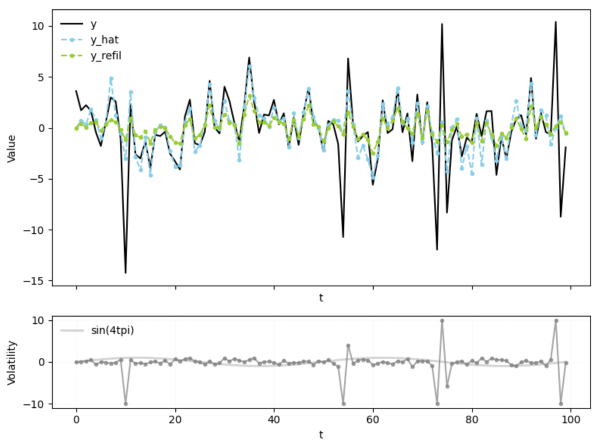


Figure 2.3: Point estimates y_hat, y_{refil} against true y.

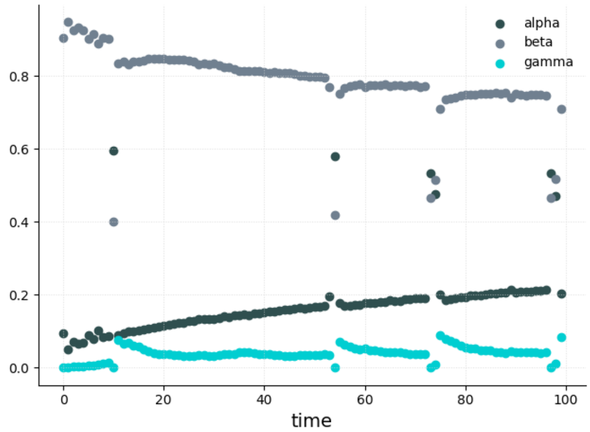


Figure 2.4: Evolution of forgetting weights.

Notice that our adaptive forgetting reacts to extremes in volatility spikes, Fig. 2.3. Initially, β is high, in order to adapt to the response. New information is assessed valuable by the model at first. A significant observation y_t is reflected by lower β and higher α, in comparison to the chosen priors β_0, γ_0. This behavior could be interpreted as the model trying to forget extreme values and retain previous information through higher α, as it only enters the weight of the previously collected statistics $\mathbf{V}_{t-1}$. After these unexpected shock, the model increased γ, the regularization parameter. This allows the model to either filter out outliers or to adapt to changing environment. In this example setting α increase over time, meaning the underlying dynamics are simple and its quickly learned.

2.4 Optimal Prior

For the inspected model (2.1), the conjugate prior is the Gaussian-inverse-Wishart (GiW) pdf. The prior pdf is specified by parameters $(\mathbf{V}_0, \Delta_0)$, where $\Delta_0 > \rho$ is the prior degrees of freedom and $\mathbf{V}_0 > 0$ is a prior extended information matrix. The matrix $\mathbf{V}_0$ has the structure of a sufficient statistic matrix (2.11),

$$\mathbf{V}_0 = \begin{pmatrix} \mathbf{V}_{f,0} & \mathbf{V}_{xf,0}^\top \\ \mathbf{V}_{xf,0} & \mathbf{V}_{x,0} \end{pmatrix}.$$

These parameters of the prior pdf must ensure the resulting prior is proper. Their choice strongly influences the structure estimation, Section 2.5, as well as forgetting, Section 2.3. The core idea, from the paper [19], is to find an optimal choice $(\mathbf{V}_0, \Delta_0)$, by maximizing marginal likelihood

$$l(\mathcal{D}^{(\bar{t})} \mid \mathbf{V}_0, \Delta_0) := \int_\Theta p(\mathcal{D}^{(\bar{t})} \mid \theta) p(\theta \mid \mathbf{V}_0, \Delta_0) d\theta$$

of the data $\mathcal{D}^{(\bar{t})}$ used for their choice. Data used to find this optimal prior is collected up to time $\bar{t}$. Thus

$$(\mathbf{V}_0, \Delta_0) \in \arg\max l(\mathcal{D}^{(\bar{t})} \mid \mathbf{V}_0, \Delta_0),$$

subject to a weak constraint that the prior is normalizable. The mentioned paper proves the existence of this extreme and gives its form. The following text summarizes the steps to compute this optimal prior.

Collect the data into the extended information matrix with a zero prior

$$\tilde{\mathbf{V}}_{\bar{t}} = \sum_{\tau=1}^{\bar{t}} \begin{pmatrix} f_\tau \\ x_\tau \end{pmatrix} \begin{pmatrix} f_\tau^\top & x_\tau^\top \end{pmatrix} + \mathbf{O} = \tilde{\mathbf{G}}_{\bar{t}} \tilde{\mathbf{G}}_{\bar{t}}^\top. \tag{2.30}$$

This gives a data-based sufficient statistic matrix. Then, select a scalar tuning parameter $\mu \in (0,1)$. This parameter controls the shrinkage strength of the resulting prior. A value around $\mu = 0.5$ is a reasonable starting point.

The optimal prior degrees of freedom, Δ_0, is then found by numerically solving the following non-linear scalar equation

$$-\frac{1}{2}\ln[\mu] = \frac{\frac{1}{2}(1-\mu)\bar{t}}{\bar{t} + (1-\mu)\Delta_0} + h(\Delta_0) - h(\bar{t} + \Delta_0), \tag{2.31}$$

where $\bar{t}$ is the number of processed data, and $h(\Delta)$ is a function involving the digamma function $\Psi(z) = \frac{d}{dz}\ln[\Gamma(z)]$, $z > 0$

$$h(\Delta) := \ln\left[\frac{1}{2}\Delta\right] - \Psi\left(\frac{1}{2}\Delta\right). \tag{2.32}$$

Equation (2.31) must be solved for $\Delta_0 > \rho$ using a numerical root-finding algorithm. The paper [19] shows that a unique solution exists.

With the value of Δ_0, the optimal prior matrix $\mathbf{V}_0$ is constructed analytically from the data-based statistics $\tilde{\mathbf{V}}$ and μ

$$\mathbf{V}_0 = \begin{pmatrix} \frac{\mu\Delta_0}{\bar{t}+(1-\mu)\Delta_0}\tilde{\mathbf{V}}_f + \frac{\mu\bar{t}}{\bar{t}(1-\mu)+(1-\mu)^2\Delta_0}\tilde{\mathbf{V}}_{xf}^\top \bar{\mathbf{V}}_x^{-1}\tilde{\mathbf{V}}_{xf} & \frac{\mu}{1-\mu}\tilde{\mathbf{V}}_{xf}^\top \\ \frac{\mu}{1-\mu}\tilde{\mathbf{V}}_{xf} & \frac{\mu}{1-\mu}\tilde{\mathbf{V}}_x \end{pmatrix}. \tag{2.33}$$

The pair $(\mathbf{V}_0, \Delta_0)$ defines the optimized conjugate prior for the linear regression problem.

These results are derived by maximizing the log-marginal likelihood function, which for the GiW prior, omitting constant terms, is:

$$l(\mathcal{D}^{(\bar{t})} \mid \mathbf{V}_0, \Delta_0) = \ln[\mathrm{J}(\tilde{\mathbf{V}}_{\bar{t}} + \mathbf{V}_0, \Delta_0 + \bar{t})] - \ln[\mathrm{J}(\mathbf{V}_0, \Delta_0)] \tag{2.34}$$

where $\mathrm{J}(\cdot)$ is the normalization constant of the GiW distribution.

Differentiating the log-likelihood with respect to Δ_0 and setting it to zero yields the key scalar equation (2.31). The structure for $\mathbf{V}_0$ in (2.33) is derived from the optimality conditions. The described choice of such prior $(\mathbf{V}_0, \Delta_0)$ makes both structure and parameter estimation more efficient, as experiments confirm.

The prior derived in this way is intuitively interpreted as a shrinkage prior. It is equivalent to setting the prior point estimates (2.13), (2.14) to

$$\hat{\mathbf{P}}_0 = \tilde{\mathbf{V}}_{x,\bar{t}}^{-1}\tilde{\mathbf{V}}_{xf,\bar{t}},$$

$$\mathbf{\Lambda}_0 = \frac{\mu\Delta_0}{\bar{t}+(1-\mu)\Delta_0}\left(\tilde{\mathbf{V}}_{f,\bar{t}} - \tilde{\mathbf{V}}_{xf,\bar{t}}^\top \tilde{\mathbf{V}}_{x,\bar{t}}^{-1}\tilde{\mathbf{V}}_{xf,\bar{t}}\right).$$

This is a profound result. The method effectively sets the prior mean of coefficients as centered exactly on the data-based point estimate. The prior beliefs about the uncertainty of the coefficients and the error variance are scaled versions of the data-based estimates. The parameter μ acts as a shrinkage factor. A smaller μ implies a weaker prior that has a larger variance (less precision) than the posterior, effectively down-weighting the prior's influence relative to the data.

The submatrices of the decomposed prior $\mathbf{G}_0\mathbf{G}_0^\top$, given by $\tilde{\mathbf{V}}_{\bar{t}} = \tilde{\mathbf{G}}_{\bar{t}}\tilde{\mathbf{G}}_{\bar{t}}^\top$ are

$$\mathbf{G}_{f,0} = \sqrt{\frac{\mu\Delta_0}{\bar{t}+(1-\mu)\Delta_0}}\tilde{\mathbf{G}}_f, \quad \mathbf{G}_{xf,0} = \sqrt{\frac{\mu}{1-\mu}}\tilde{\mathbf{G}}_{xf}, \quad \mathbf{G}_{x,0} = \sqrt{\frac{\mu}{1-\mu}}\tilde{\mathbf{G}}_x.$$

These matrices don't need to be recalculated, unless the user recognizes that the underlying system has changed significantly. Then the choice of the prior statistics, as well as structure estimation, Section 2.5 are repeated, utilizing the collected data $\mathbf{G}_t, \Delta_t$. For practical use, begin with very small diagonal matrix $\approx 10^{-6}\mathbf{I}$, instead of the zero matrix in (2.30), which ensures numerical stability. Algorithm 1, given in Section 2.3, serves to to construct $\tilde{\mathbf{G}}_{\bar{t}}$.

2.5 Structure Estimation

Estimating a suitable structure for the ARX model (2.1) is a crucial task in overall regression problems. We assume ρ available regressors/predictors that are potentially significant for predicting y_t. Selecting an appropriate subset of these into the appropriate x_t, from all considered entries in $\tilde{x}_t$ is computationally demanding, as there are $2^\rho - 1$ combinations of including and excluding regressors, not accounting for a zero vector $\tilde{x}_t = \mathbf{0}$. In frequentist statistics, this estimation is usually done by performing a stepwise regression, comparing the models based on the Bayesian information criterion (BIC) or the Akaike's information criterion (AIC), [2]. Their approximate nature limits their performance. An efficient, ideally non-approximate, Bayesian structure estimation method should be used. The Chapter [22] offers such a maximum a posteriori probability estimator.

The likelihood space in our case is highly non-convex and multi-modal, implying that the combination of regressors, which leads to a higher likelihood is not trivial. For the purpose of finding a probable global maxima, we used a Genetic Algorithm (GA), [3], that promotes exploration over exploitation. Algorithm 2 provides its pseudo-code. The marginal log-likelihood to be evaluated is

$$\begin{aligned}\ln l(m) \propto \sum_{j=1}^{N} &\ln\Gamma(\tfrac{\Delta_t-\rho_m+N+1-j}{2}) - \ln\Gamma(\tfrac{\Delta_0-\rho_m+N+1-j}{2}) - \tfrac{N}{2}(\ln|\mathbf{V}_{x,t}(m)| - \ln|\mathbf{V}_{x,0}(m)|) \\ &- \tfrac{\Delta_0+t-\rho_m+N}{2}\ln|\mathbf{\Lambda}_t(m)| + \tfrac{\Delta_0-\rho_m+N}{2}\ln|\mathbf{\Lambda}_0(m)|\end{aligned} \tag{2.35}$$

There, m signifies the dependence on the examined possible structure, $t > \bar{t}$, see Section 2.4. The ρ-vector m contains units at those positions of $\tilde{x}_t$, which appear in the currently considered x_t and zeros in all other positions. ρ_m is the number of ones in the examined vector m and $\mathbf{V}(m)$ is a matrix gained from $\mathbf{V}(m = \mathbf{1})$. It omits rows and columns, corresponding to zeros in the vector m. $\boldsymbol{\Lambda}(m) = \mathbf{V}_f - \mathbf{V}_{xf}^{\top}(m)\mathbf{V}_x^{-1}(m)\mathbf{V}_{xf}(m)$, from (2.14), where in $\mathbf{V}_{xf}(m)$ only rows are omitted. The prior $\mathbf{V}_0, \Delta_0$ from Section 2.4 was substituted into the log-likelihood (2.35). The square-root version of the extended information matrix (2.19) makes the evaluation of determinants in (2.35) computationally safe and cheap.

Algorithm 2 Genetic Pseudo-Algorithm for Regressor Subset Selection

Require: Initial parent vector $\blacksquare = [0, 0, \ldots, 0]$, $\mathrm{max}_{\mathrm{iter}}$, decay rate alpha $\in (0, 1)$, mutation probability $\mathrm{p}_{\mathrm{mut}} \in (0, 1)$, batch-size

```
ln l_max ← −inf
iter ← 0
while iter < max_iter do
    batch = generate_batch(batch-size, p_mut, ■)          ▷ see below
    for m in batch do
        Evaluate ln l(m)                                   ▷ according to (2.35)
    end for
    best_m ← arg max ln l(m)                               ▷ select the best mutation
    if ln l(best_m) > ln l_max then
        ■ ← best_m
        ln l_max ← ln l(best_m)
    end if
    offspring = crossover(best_m, ■)                       ▷ see below
    if ln l(offspring) > ln l_max then
        ■ ← offspring
        ln l_max ← ln l(offspring)
    end if
    p_mut ← alpha · p_mut
    iter ← iter + 1
end while
return (■, ln l_max)
```

Genetic algorithms, broadly, are an optimization method inspired by natural evolution. They initiate with a population of candidates, evaluate their fitness by a user defined function, here (2.35), and iteratively improve the candidates through systematic selection, followed by a crossover (recombination of good population members) and mutation (introducing small random changes). Over successive generations, the population evolves towards a better solution, making genetic algorithms well-suited for complex or poorly understood search spaces, precisely like in our case, where small changes in the used regressors x_t may have large effects on the predictive accuracy.

In our case, the population consists of vectors $\{m\}$ of length ρ, comprising ones and zeros. As stated previously, ones correspond to the regressors included into the model and zeros correspond to the excluded regressors.

Mutation randomly flips each bit in m with a certain probability $\mathrm{p}_{\mathrm{mut}}$.

The offspring is created by a crossover of two parents via single-point crossover method. An integer $k < \rho$ is selected with uniform probability over values $\{1, \ldots, \rho - 1\}$. The first k bits are contributed by the first parent, and the next $\rho - k$ bits are contributed by the second one.

The best mutated vector in a given population may be selected via a roulette-wheel selection process based on the probabilities assigned by soft-max transformation of their respective log-likelihoods,

$$\frac{\exp[\ln l(m)]}{\sum_{\{m\}} \exp[\ln l(m)]},$$

or by elitism selection, which was chosen in our implementation. This selects the vector m with the highest log-likelihood among the current population as the new parent $\blacksquare$, which goes into the next iteration.

A good choice of parameters for the Algorithm 2 is problem specific. Basic recommendation is given. The maximum number of iterations should scale linearly with ρ, initial mutation probability $\mathrm{p_{mut}} \in (0.1, 0.5)$, and a value around 0.25 should be satisfactory for most problems. The decay rate alpha is set so that the mutation probability is not effectively 0 after a large number of iterations. The Algorithm is presented to start at a zero vector, modeling independent returns. Thought, for very large ρ, we can utilize multiple populations with randomly initialized parents $(\blacksquare_1, \dots \blacksquare_n)$, and seek the maximum likelihood across those runs.

Example 2.2.

Take $\rho = 16, \bar{t} = 30, t = 100, \mu = 0.5$[9], the true number of regressors used to generate the scalar response ($N = 1$), is 5. Data was generated randomly with seed 42 in numpy, similarly to Example 2.1, $y_t = \mathbf{P}^\top x_t + e_t$, $e_t \sim \mathcal{N}(0, 0.2^2)$, $x_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, $\mathbf{P}$ is a ρ-vector with five non-zero entries. The number of possible combinations of regressors, including the zero model, is $2^{16} = 65536$. The selected parameters for the Algorithm 2 are $\mathrm{p_{mut}} = 0.25, \text{alpha} = 0.89, \text{batch}_{\text{size}} = 2, \max_{\text{iter}} = 50$. The maximum number of evaluated possibilities is reduced to $150 = \max_{\text{iter}}(\text{batch}_{\text{size}} + \#\text{offsprings})$. The evolution of the population is shown below, in Figs. 2.5, 2.6. The results achieved with prior $(\mathbf{V}_0, \Delta_0)$, as described in Section 2.4, are compared to the prior $(10^{-4}\mathbf{I}, \bar{t})$. Regressors that were truly used to generate the data are highlighted with an red outline.

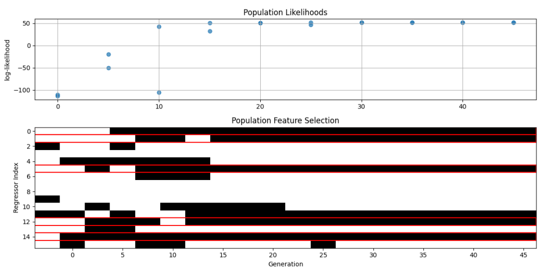


Figure 2.5: Genetic Algorithm with our prior.

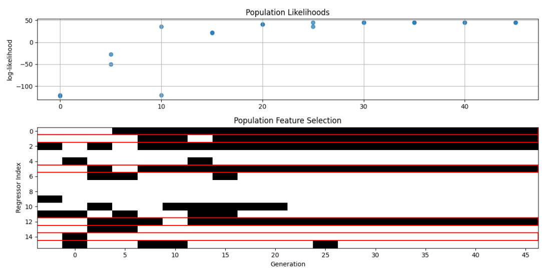


Figure 2.6: Genetic Algorithm with non-optimized prior.

Our approach to the problem of structure estimation is compared with a standard stepwise regression based on the AIC limited to 150 steps. The result is shown in Fig. 2.7. Although the stepwise AIC estimation, in this particular example, correctly identified all of the true regressors,

[9] The optimal prior $\mathbf{V}_0$ depends on μ, the recommended values are $\mu \in (0.5, 0.95)$.

the number of retained regressors is large compared to the our way. It corresponds with the fact that AIC does not reflect covariance of parameter estimates.

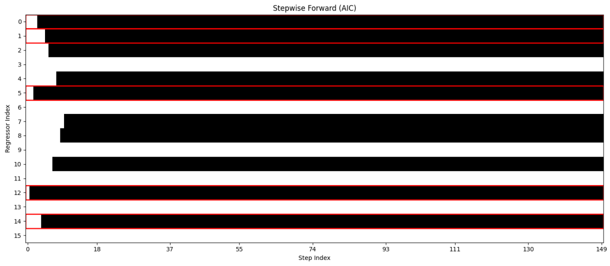


Figure 2.7: Significant regressors according to stepwise regression based on AIC.

3 Fully Probabilistic Design

3.1 Formalism

The Fully Probabilistic Design (FPD) [17] of decision strategies provides a unified, axiomatic framework for decision-making under uncertainty. It generalizes classical methods such as Markov Decision Process (MDP) and Linear-Quadratic Regulator (LQR) control by formulating them in terms of an *ideal* probability distribution over the entire system behavior. The optimal decision strategy is defined as the one whose induced distribution over actual behavior is the closest to this ideal distribution. Their proximity is measured by KLD. This formulation naturally yields randomized policies, which offer an inherent mechanism for balancing exploration and exploitation. Importantly, it helps us to quantify user's aims, including risk aversion.

The FPD focuses on closed-loop interaction between an agent and their environment.

At discrete time steps $t \in \{1, 2, ...\}$, the agent observes states s_{t-1} and takes actions $a_t \in \mathcal{A}$. Then the environment transitions to state s_t. This interaction is structurally identical to that considered in MDP.

The behavior b is defined as the collection of all relevant random variables up to the decision horizon h, including actions, states and any unknown parameters $\theta \in \Theta$ of the environment model[10],

$$b = (s_h, a_h, s_{h-1}, a_{h-1}, ..., s_1, a_1, s_0, \theta).$$

The joint probability density function of the (closed-loop) behavior is

$$c_r(b) := \left(\prod_{t=1}^{h} m(s_t \mid s_{t-1}, a_t, \theta) \right) \left(\prod_{t=1}^{h} r(a_t \mid s_{t-1}) \right) p(\theta)$$

and represents the response to a given strategy $r = (r(a_t \mid s_{t-1})_{t-1}^{h}$. Here, the pdf $m(s_t \mid s_{t-1}, a_t, \theta)$ describes the environment's dynamics, the pdf $r(a_t \mid s_{t-1})$ represents the causal, generally randomized decision rules defining the agent's strategy that meets the natural conditions of control [28], and $p(\theta)$ is the prior pdf over the unknown parameters θ.

The agent specifies its objectives and constraints through an *ideal* joint pdf $c^I(b)$ over behaviors. This ideal assigns high probability to desirable behaviors and low or zero probability to

[10] The ARX model can always be converted into state-space via a suitable state definition. We exploit this possibility here, in order to provide a concise description of FPD and its LQR version, see Sec. 3.3

undesirable or forbidden ones. It is factorized analogously to $c_r(b)$.

The FPD-optimal decision strategy minimizes the KLD

$$\min_{(r(a_t|s_{t-1}))_{t=1}^h} D(c_r(b)||c(b)^I) = \min_{(r(a_t|s_{t-1}))_{t=1}^h} \int c_r(b) \ln\left[\frac{c_r(b)}{c^I(b)}\right] db. \tag{3.1}$$

The KLD in (3.1) can be written as the expected cumulative loss,

$$D(c_r||c^I) = \mathrm{E}_{c_r}\left[\ln\left[\frac{c_r(b)}{c^I(b)}\right]\right] = \mathrm{E}_{c_r}\left[\sum_{t=1}^h L_t(s_t, s_{t-1}, a_t, \theta)\right],$$

where the loss at time t is

$$L_t = \ln\left[\frac{m(s_t \mid s_{t-1}, a_t, \theta) r(a_t \mid s_{t-1})}{m^I(s_t \mid s_{t-1}, a_t, \theta) r^I(a_t \mid s_{t-1}, \theta)}\right]. \tag{3.2}$$

The optimization problem (3.1) is solved via dynamic programming, i.e. backwards recursion from the terminal time horizon h to 1, or $t+h$ to t, specified as per context. Let $v(s_t) \geq 0$ denote the optimal expected future loss from state s_t, with the terminal condition $v(s_h) = 0$.

For convenience, value function $v(s_{t-1})$ is expressed as

$$v(s_{t-1}) = -\ln[n_t(s_{t-1})], \quad n_t(s_{t-1}) \in [0, 1].$$

The core part of the recursion at step t is to find the rule $r(a_t \mid s_{t-1})$ that minimizes the expected current-plus-optimal-future loss. This leads to the following recursive equations:

1. Compute an auxiliary function $d_t(a_t, s_{t-1})$

$$:= \int_{\mathcal{S},\Theta} p(\theta \mid s_{t-1}) m(s_t \mid s_{t-1}, a_t, \theta) \left(\ln\left[\frac{m(s_t \mid s_{t-1}, a_t, \theta)}{m^I(s_t \mid s_{t-1}, a_t, \theta) r^I(a_t \mid s_{t-1}, \theta)}\right] + v(s_t)\right) ds_t d\theta.$$

With $v(s_t) = -\ln[n_{t+1}(s_t)]$, it becomes

$$d(s_{t-1}, a_t) = \mathrm{E}\left[\ln\left[\frac{m(s_t \mid s_{t-1}, a_t, \theta)}{n_{t+1}(s_t) m^I(s_t \mid s_{t-1}, a_t, \theta) r^I(a_t \mid s_{t-1}, \theta)}\right] \mid s_{t-1}, a_t\right]. \tag{3.3}$$

2. The minimizing rule $r^o(a_t \mid s_{t-1})$ takes the exponential form

$$r^o(a_t|s_{t-1}) = \frac{\exp[-d_t(s_{t-1}, a_t)]}{\int_{\mathcal{A}} \exp[-d_t(s_{t-1}, a')] da'} \propto \exp[-d_t(s_{t-1}, a_t)].$$

3. Update the value function $v(s_{t-1})$ for the previous time

$$v(s_{t-1}) = -\ln[n_t(s_{t-1})],$$
$$n_t(s_{t-1}) = \int_{\mathcal{A}} \exp[-d_t(s_{t-1}, a_t)] da_t.$$

As previously noted, the recursion is initiated with $v(s_h) = 0$, equivalently $n_{h+1}(s_h) = 1$ and proceeds backwards to $t = 1$. The resulting collection of rules $(r^o(a_t \mid s_{t-1}))_{t=1}^h$ constitutes the FPD-optimal strategy. A noteworthy property of this construction is that it yields a randomized (soft-max-style) strategy, which is essential for systematic exploration.

3.2 Exploration

Exploration is a crucial part of strategy design if the agent exerts influence over the system. Though we assumed our actions do not influence the market, exploration in general could be beneficial to large investors. In our case, stochasticity in actions is not used to explore the system but rather to acknowledge uncertainty in the deterministic optimal actions, due to possible improper selection of the system's parameters' point estimates.

We examine non-deterministic strategies and how to recover them via the FPD from a proper selection of the *ideal* pdf.

The idea of Boltzmann sampling [32] or simulated annealing is to control the strategy with a hyperparameter T, called temperature-controlled sampling,

$$\left(r^o(a_t \mid s_{t-1})\right)^{1/T}.$$

- $T \to 0$: selects the mode of $r^o(a_t \mid s_{t-1})$ (pure exploitation),
- $T = 1$: uses $r^o(a_t \mid s_{t-1})$,
- $T \to \infty$: approaches uniform distribution (pure exploration).

Usually, one would start with a high T[11] to promote exploration during the learning phase and gradually lower it to transition to an exploitation phase as confidence in agent's action rises. This sampling is mostly seen as a trick in training, to push the agent to try different actions and learn the system response.

We show that this can arise in the FPD formalism from a higher level principle. Assume an agent, that defines their *ideal* closed-loop model as the geometric mean of the actual $c_r(b)$ and a base *ideal* pdf $c^I(b)$. Then the loss (3.2) is

$$\ln\left[\frac{c_r(b)}{[c^I(b)]^{\frac{1}{T}} c_r(b)^{1-\frac{1}{T}}}\right] = \frac{1}{T}\ln\left[\frac{c_r(b)}{c^I(b)}\right].$$

The auxiliary function (3.3) then becomes $d_t(a_t, s_{t-1}) = \mathrm{E}[\frac{1}{T}L_t | s_{t-1}, a_t]$, see (3.2), leading to *Boltzmann agent*. This aligns with the description above. With T closer to 0 the agent is more confident about the actual $c_t(b)$ being closer to their *ideal* and wants to concentrate their actions around the mean for exploitation, while the agent's growing uncertainty about their *ideal* is reflected in higher T values. The gained decision strategy is proportional to the exponential of the auxiliary function $d(s_{t-1}, a_t)$, divided by T

$$r^o(a_t \mid s_{t-1}, T) \propto \exp\left[-\frac{1}{T} d(s_{t-1}, a_t)\right] = \left(\exp\left[-d(s_{t-1}, a_t)\right]\right)^{\frac{1}{T}} = \left(r^o(a_t, s_{t-1})\right)^{\frac{1}{T}}.$$

Thus, we have successfully recovered the Boltzmann sampling from FPD by clearly expressing the agent's interpretation of the ideal pdf.

Another FPD-compatible strategy is Thompson sampling [31], which exploits the Bayesian

[11] T is another hyperparameter to be optimized or selected by an expert. Also, its potential evolution is to be selected

structure of the system model by sampling parameter matrices $\tilde{\mathbf{P}}$ directly from its posterior pdf (2.16)

$$\tilde{\mathbf{P}} \sim p(\mathbf{P} \mid \mathcal{D}^{(t)}, \tilde{\mathbf{\Sigma}}),$$

with $\tilde{\mathbf{\Sigma}}$ sampled from $p(\mathbf{\Sigma} \mid \mathcal{D}^{(t)})$. Thompson sampling could be particularly beneficial in the backwards Riccati recursion, Chapter 4, as it allows uncertainty in parameter estimates to be elegantly incorporated into the decision-making process. Suppose an system parameterized by unknown matrices $\theta = (\mathbf{P}, \mathbf{\Sigma})$ with posterior $p(\theta \mid \mathcal{D}^{(t)})$, given the data $\mathcal{D}^{(t)}$ up to time t. In the exact FPD setup, the optimal strategy is

$$r^o(a_t \mid s_{t-1}) \propto \exp[-d(s_{t-1}, a_t)].$$

The induced auxiliary function becomes

$$d(s_{t-1}, a_t) = \mathrm{E}_{\theta \sim p(\theta | \mathcal{D}^{(t)})} \left[d_\theta(s_{t-1}, a_t) \right],$$

where

$$d_\theta(s_{t-1}, a_t) = \int_{\mathcal{S}} m(s_t \mid s_{t-1}, a_t, \theta) \ln \left[\frac{m(s_t \mid s_{t-1}, a_t, \theta)}{n_{t+1}(s_t) m^I(s_t \mid s_{t-1}, a_t, \theta) r^I(a_t \mid s_{t-1}, \theta)} + v(s_t) \right] ds_t$$

To approximate this expectation, we sample and set parameters $\tilde{\theta} \sim p(\theta \mid \mathcal{D}^{(t)})$ and then use the optimal strategy under that sample

$$\hat{d}_t(a_t, s_{t-1}) = d_{\tilde{\theta}}(a_t, s_{t-1})$$

Practically, this corresponds to the Thompson sampling rule, where we draw $\tilde{\theta} \sim p(\theta \mid \mathcal{D}^{(t)})$, compute the optimal strategy $r^o_{\tilde{\theta}}$ under the sampled parameters and set $d_{\tilde{\theta}}$ in dynamic programming. The description is valid for any case, not only LQR. Thus, Thompson sampling is recovered as the FPD-optimal strategy under parameter uncertainty, with randomness arising not from the action space itself but from the Bayesian posterior and approximate evaluation of its expectation. This provides a meaningful approximate dynamic programming, [12], [31].

3.3 Linear Quadratic Regulator

We demonstrate how the classic LQR emerges as a special case of FPD. We use this relation for a justified quantification of user's aims, including their aversion to risk. Consider a linear system with Gaussian noise,

$$s_t = \mathbf{A}_{t-1} s_{t-1} + \mathbf{B} a_t + \epsilon_t, \quad \epsilon \sim \mathcal{N}(0, \mathbf{\Sigma}_\epsilon), \tag{3.4}$$

with the state vector s_t made of actions a_t, a_{t-1}, corresponding observations y_t, regressors and constant term 1,

$$s_t = \begin{pmatrix} a_t \\ a_{t-1} \\ y_t \\ x_t \\ 1 \end{pmatrix}, \quad s_t \in \mathbb{R}^{3N+\rho+1}. \tag{3.5}$$

The evolution matrices $\mathbf{A}_{t-1}, \mathbf{B}$ in (3.4) take the form[12]

$$\mathbf{A}_{t-1} = \begin{pmatrix} \mathbf{O} & \mathbf{O} & \dots & \dots & 0 \\ \mathbf{I} & \mathbf{O} & \dots & \dots & 0 \\ \mathbf{O} & \mathbf{O} & \dots & \hat{\mathbf{P}}_{t-1} & 0 \\ \mathbf{O} & \mathbf{O} & \hat{\mathbf{X}}_{t-1} & \mathbf{O} & 1 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} \mathbf{I} \\ \mathbf{O} \\ \vdots \\ 0^\top \end{pmatrix}, \tag{3.6}$$

$\mathbf{A}_{t-1} \in \mathbb{R}^{3N+\rho+1, 3N+\rho+1}$, $\mathbf{B} \in \mathbb{R}^{3N+\rho+1, N}$, and $\hat{\mathbf{X}}_{t-1} \in \mathbb{R}^{N+\rho+1, \rho+1}$ is the matrix that approximates the next regressors $\hat{\mathbf{X}}_{t-1} x_t = \hat{x} \approx x_t$, while the corresponding, zero-mean innovation e_x has covariance $\mathbf{\Sigma}_x$. Often, $\hat{\mathbf{X}}_{t-1}$ can be chosen as a constant matrix realizing shift in x_{t-1} and new parts are modeled as a random walk. The covariance matrix in (3.4) is $\hat{\mathbf{\Sigma}}_\epsilon = \mathrm{diag}(\mathbf{O}, \mathbf{O}, \hat{\mathbf{\Sigma}}, \hat{\mathbf{\Sigma}}_x, 0)$, where $\hat{\mathbf{\Sigma}} = \hat{\mathbf{\Sigma}}_{t-1}$ is given by (2.21) ($\hat{\mathbf{\Sigma}}_\varepsilon$ is time-varying).

If the constant term is part of x, then the size of the matrix changes appropriately. Depending on the structure estimation, Section 2.5, x_t may already the constant term, in which case it need not to be listed explicitly.

Let us stress that the expected state vector $\hat{s}_t$ contains the expectation $\hat{y}_t$ given by $\hat{\mathbf{P}}_{t-1}^\top x_t$ and also the expected regressor $\hat{x}_t$. The variables in the regressor also need their own, at least, simple model. It is also assumed to be Gaussian with expectation $\hat{\mathbf{X}}_{t-1} x_t$ and covariance $\hat{\mathbf{\Sigma}}_x$.

3.3.1 LQR as a special case of FPD

The classical LQR objective is to minimize the expected cumulative quadratic loss

$$\mathrm{E}\left[\sum_{t=1}^{h} s_t^\top \mathbf{Q}_{t-1} s_t\right], \tag{3.7}$$

where $\mathbf{Q}_{t-1} \geq 0$. Within FPD, we set the ideal decision rule r^I equal to the rule being optimized ($r^I(a_t \mid s_{t-1}) = r(a_t \mid s_{t-1})$), so that the decision rule component of KLD vanishes, $\ln\left(\frac{r_t}{r_t^I}\right) = 0$. It makes no harm as a_t is a part of the state (3.5). It, however, leads to the standard deterministic strategy, see [16]. The quadratic loss is then encoded directly into the *ideal* environment model as

$$m^I(s_t \mid s_{t-1}, a_t) \propto m(s_t \mid s_{t-1}, a_t) \exp\left(-s_t^\top \mathbf{Q}_{t-1} s_t\right), \tag{3.8}$$

where the exponential term encodes our request for minimizing a quadratic loss into the pdf directly. This *ideal* pdf is Gaussian and expresses the *wish* to reach the state that minimizes our loss. With this choice, the FPD loss reduces precisely to our desired LQ loss

$$L_t = \ln\left[\frac{m(s_t \mid s_{t-1}, a_t) r(a_t \mid s_{t-1})}{m^I(s_t \mid x_{t-1}, a_t) r^I(a_t \mid s_{t-1})}\right] = \ln\left[\frac{1}{\exp[-s_t^\top \mathbf{Q}_{t-1} s_t]}\right] = s_t^\top \mathbf{Q}_{t-1} s_t.$$

Thus, minimizing the KLD, $\min_r D(c_r || c^I)$ in FPD, is equivalent to the standard LQR objective

$$\min_{(a_i)_{i=1}^h} \mathrm{E}\left[\sum_{t=1}^{h} s_t^\top \mathbf{Q}_{t-1} s_t\right].$$

[12] We exploit certainty-equivalence approximation [15] by replacing unknown parameters with their newest point estimates, gained via recursive Bayesian learning. It is computationally much cheaper than the discussed sampling.

3.3.2 From FPD to Risk-Averse Design

The LQR formulation so far is risk-ignorant, in the sense that it optimizes the expected future returns without considering their volatility. In financial applications, however, most investors prefer portfolios with uniform, ideally low volatility across holdings, even at the expense of smaller average returns. Traditionally, this is modeled in Markowitz's mean-variance [26] dual objective optimization, where the objective is to maximize expected return and minimize the expected variance. Symbolically,

$$\max_a \left(\mathrm{E}[\text{Returns}] - \omega \mathrm{Var}[\text{Returns}]\right),$$

with $\omega \geq 0$ quantifying the degree of individual investors' risk aversion.

We show how the risk-averse strategy emerges naturally by carefully selecting our *ideal* pd that align with our goals and how to select the parameter ω. Let us only work with the parts of the state that are important for the objective of portfolio optimization. Those are the vector of returns y, action a and past actions a_{-1}, where we temporarily dropped the time index t for simplicity. The goal of the investor is to maximize their expected return $a^\top y$, see Fig. 1.1, and minimize the transaction costs given by $\mathbf{D}$, see Chapter 1. The pdf of the response variable y is

$$m(y \mid a) = m(y) \propto \mathcal{N}(\hat{y}, \mathbf{\Sigma}), \quad \hat{y} = \hat{\mathbf{P}}x,$$

since, for the examined problem, the investor has no influence on the markets[13]. Suppose that we are unsure about what the *ideal* pd $c^I(y, a) = m^I(y \mid a) r^I(a)$ should look like. The ideals are defined as to meet our aims to minimize transaction costs and maximize returns, while acknowledging uncertainty. The *ideal* pdfs that reflect this are taken as

$$r^I(a) \sim \mathcal{N}(a_{-1}, \mathbf{D}^{-1}), \quad m^I(y \mid a) \propto m(y) \exp\Big[\omega a^\top y\Big], \quad \omega \geq 0,$$

The pdf $m^I(y \mid a)$ is treated as if y depends on a because in the *ideal* case, the agent chooses its action and then the return y aligns with a to maximize $a^\top y$, which holds if and only if $y = y(a) = \omega a$, for some $\omega \geq 0$. The *ideal* risk-averse m is then found by choosing r and ω as optimizers in

$$\max_{\omega > 0} \min_r D(c_r || c^I). \tag{3.9}$$

Recall the auxiliary function $d(a_t, s_{t-1})$, (3.3), which in the used simplified notation reads

$$d(a) = \mathrm{E}_y \left[\ln\left[\frac{m(y)}{m^I(y \mid a)}\right]\right] - \ln\left[r^I(a)\right],$$

where we also omitted the remaining terms that are not involved in this particular optimization problem. Then (3.9) takes the form

$$\max_{\omega \geq 0} - \ln \int r^I(a) \exp\left[-\mathrm{E}_y\left[\ln\left[\frac{m(y)}{m^I(y \mid a)}\right]\right]\right] da. \tag{3.10}$$

We use Lemma *A.9* to complete the square in $m^I(y \mid a)$

$$\begin{aligned} m^I(y \mid a) &\propto \exp\left[-\frac{1}{2}\Big((y-\hat{y})^\top \mathbf{\Sigma}^{-1}(y-\hat{y}) - 2\omega a^\top y\Big)\right] \\ &\propto \exp\left[-\frac{1}{2}\Big((y-\hat{y}-\omega\mathbf{\Sigma}a)^\top \mathbf{\Sigma}^{-1}(y-\hat{y}-\omega\mathbf{\Sigma}a) - 2\omega a^\top \hat{y} - \omega^2 a^\top \mathbf{\Sigma} a\Big)\right] \\ &\sim \mathcal{N}(\hat{y} + \omega\mathbf{\Sigma}a, \mathbf{\Sigma}). \end{aligned}$$

[13] For sufficiently large investors, it is not hard to make y dependent on a, through $x = x(a)$.

Then $\mathrm{E}_y\left[\ln\left[\frac{m(y)}{m^I(y|a)}\right]\right]$ is the KLD of two Gaussian distributions with identical covariance,

$$D(\mathcal{N}(\hat{y},\mathbf{\Sigma})||\mathcal{N}(\hat{y}+\omega\mathbf{\Sigma}a,\mathbf{\Sigma})) = \frac{\omega^2}{2}a^\top\mathbf{\Sigma}a.$$

Putting this back into (3.10) and substituting for $r^I(a)$ gives

$$\max_{\omega\geq 0}\int \exp\left[-\frac{1}{2}\left((a-a_{-1})^\top\mathbf{D}(a-a_{-1})+\omega^2a^\top\mathbf{\Sigma}a\right)\right]da,$$

after applying Lemma *A.9*, the integral to be maximized yields

$$\int \exp\left[-\frac{1}{2}\left((a-\mathbf{E}^{-1}\mathbf{D}a_{-1})^\top\mathbf{E}(a-\mathbf{E}^{-1}\mathbf{D}a_{-1})+a_{-1}^\top(\mathbf{D}-\mathbf{D}\mathbf{E}^{-1}\mathbf{D})a_{-1}\right)\right]da,$$

where $\mathbf{E}=\mathbf{D}+\omega^2\mathbf{\Sigma}$. This is a Gaussian integral that can be evaluated, and after applying the negative logarithm and dropping the constant, (3.10) takes the form

$$\frac{1}{2}\ln|\mathbf{E}|+\frac{1}{2}a_{-1}^\top(\mathbf{D}-\mathbf{D}\mathbf{E}^{-1}\mathbf{D})a_{-1},$$

differentiating with respect to ω gives the necessary condition for extreme gets

$$\omega\mathrm{tr}\left(\mathbf{E}^{-1}\mathbf{\Sigma}\right)+\omega a_{-1}^\top\mathbf{D}\mathbf{E}^{-1}\mathbf{\Sigma}\mathbf{E}^{-1}\mathbf{D}a_{-1}=0.$$

A direct inspection of the optimized function implies that its solution is a maximizer. The optimal choice for ω can be found by a simple root-finding algorithm. With ω chosen, the loss (3.2) to be evaluated is

$$L=-\omega a^\top\hat{y}+(a-a_{-1})^\top\mathbf{D}(a-a_{-1})+\frac{\omega^2}{2}a^\top\mathbf{\Sigma}a,$$

where we have to use the point estimates for $\hat{y}$ and $\mathbf{\Sigma}$, see Section 2.1, $\hat{y}=\mathbf{P}^\top x$, $\hat{\mathbf{\Sigma}}=\frac{1}{\Delta}\mathbf{G}_f\mathbf{G}_f^\top$. This loss can be written as a quadratic criterion $\mathbf{Q}\geq 0$,

$$\begin{pmatrix}a^\top & a_{-1}^\top & \hat{y}^\top\end{pmatrix}\mathbf{Q}\begin{pmatrix}a\\ a_{-1}\\ \hat{y}\end{pmatrix}.$$

To describe the loss matrix $\mathbf{Q}$, we adopt simplifying expressions, $\mathbf{C}=\mathbf{D}+\frac{\omega^2}{2}\mathbf{\Sigma}$, $b=\mathbf{C}^{-1}\mathbf{D}a_{-1}$, $c=\omega\hat{y}$. Then we apply both identities from Lemma *A.9*

$$L=\left(a-b-\frac{1}{2}\mathbf{C}^{-1}c\right)^{-1}\mathbf{C}\left(a-b-\frac{1}{2}\mathbf{C}^{-1}c\right)\left\{-b^\top c-\frac{1}{4}c^\top\mathbf{C}^{-1}c-a_{-1}\mathbf{D}b+a_{-1}\mathbf{D}a_{-1}\right\},$$

where the terms in the brackets $\{\}$ do not depend on a and are, therefore, they are insignificant in the optimization for action. Then, when we stress the dependence of time,

$$\mathbf{Q}=\mathbf{Q}_t=\begin{pmatrix}\mathbf{C}_t & -\mathbf{D}_t & -\frac{\omega_t}{2}\mathbf{I}\\ -\mathbf{D}_t & \mathbf{D}_t\mathbf{C}_t^{-1}\mathbf{D}_t & \frac{\omega_t}{2}\mathbf{D}_t\mathbf{C}_t^{-1}\\ -\frac{\omega_t}{2}\mathbf{I} & \frac{\omega_t}{2}\mathbf{C}_t^{-1}\mathbf{D}_t & \frac{\omega_t^2}{4}\mathbf{C}_t^{-1}\end{pmatrix}. \tag{3.11}$$

If $\mathbf{Q}_t$ is not positive-definite, we add a small diagonal to the second and third diagonal blocks. This practically does not change the designed strategy but prevents numerical issues.

The parameter ω depends on the previous action a_{-1}, hence ω_t should be used within quadratic programming in the following Chapter, 4. Unless only a single step is performed in the optimization, as in Section 4.2, we take $\omega_t\approx\omega_\tau$.

4 Optimization Algorithm

4.1 Multi-Step Optimization

Let us be at time τ, with our known previous allocation $a_{\tau-1}$. We deal with a Constrained Sequential Quadratic Programming problem

$$v(s_{t-1}) = \min_{a_t \in \Omega_{BE}} \mathrm{E}[s_t^\top \mathbf{Q}_{t-1} s_t + v(s_t) \mid a_t, \mathcal{D}^{(t-1)}], \quad \Omega_{BE} = \{a_t \in \mathbb{R}^N \mid \mathbf{W} a_t = \mathbf{c}, a_t \geq \mathbf{l}\}. \quad (4.1)$$

In our case $\mathbf{W} = \mathbf{1}^\top$, $\mathbf{c} = 1$, $\mathbf{l} = \mathbf{0}$, so $\Omega_{BE} = \mathcal{A}$. It is assumed that the Bellman function $v(.)$, with the loss-to-go matrix $\mathbf{H}$ can be written as

$$v(s_t) = ||\mathbf{H}_t s_t||^2 + \psi_t.$$

The presented optimization method builds upon the Semi Monotonic Augmented Lagrangian algorithm for Bound and Equality constrained (SMALBE) minimization [12]. The matrix $\mathbf{H}_t$ is kept in an upper triangular form and the scalar offset ψ_t is independent of the data. The optimization process is divided into two phases, outer and inner loops.

The outer loop of the algorithm minimizes the Lagrangian, in which the scalar λ cares about the equality constraint and propagates the loss-to-go determining matrix $\mathbf{H}$ back in time. The first step is to perform the backward induction from the finite horizon h to calculate the final cost-to-go matrix, considering the equality constraint. The initial values are set to $\mathbf{H}_{\tau+h} := \mathbf{O}$, $\psi_{t+h} := 0$, [6],

$$v(s_{t-1}) = \min_{a_t} \mathrm{E}[s_t^\top (\mathbf{Q}_{t-1}^{\frac{\top}{2}} \mathbf{Q}_{t-1}^{\frac{1}{2}} + \mathbf{H}_t^\top \mathbf{H}_t) s_t \mid a_t, \mathcal{D}^{(t-1)}] + \psi_t + \lambda(\mathbf{1}^\top a_t - 1) =$$

$$= \min_{a_t} \mathrm{E}\left[\left|\left| \begin{pmatrix} \mathbf{Q}_{t-1}^{\frac{1}{2}} \\ \mathbf{H}_t \end{pmatrix} s_t \right|\right|^2 \mid a_t, \mathcal{D}^{(t-1)} \right] + \psi_t + \lambda(\mathbf{1}^\top a_t - 1).$$

The matrix $\begin{pmatrix} \mathbf{Q}_{t-1}^{\frac{1}{2}} \\ \mathbf{H}_t \end{pmatrix}$ is transformed by orthogonal transformation to an upper triangular matrix, denoted $\mathbf{H}_t^{'}$, with zeros under the end of the main diagonal, thus making it effectively a $(3N + \rho + 1, 3N + \rho + 1)$-matrix. We apply expectation and get the following, see (3.4),

$$v(s_{t-1}) = \min_{a_t} \left|\left| \mathbf{H}_t^{'} \begin{pmatrix} \mathbf{B} & \mathbf{A}_{t-1} \end{pmatrix} \begin{pmatrix} a_t \\ s_{t-1} \end{pmatrix} \right|\right|^2 + \mathrm{tr}(\mathbf{H}_t^{'\top} \mathbf{H}_t^{'} \mathbf{\Sigma}_\epsilon^{\frac{\top}{2}} \mathbf{\Sigma}_\epsilon^{\frac{1}{2}}) + \psi_t + \lambda(\mathbf{1}^\top a_t - 1).$$

Applying orthogonal transformation again, the resulting matrix under the norm becomes

$$\tilde{\mathbf{H}}_t = \begin{pmatrix} \tilde{\mathbf{H}}_{t,a} & \tilde{\mathbf{H}}_{t,s} \\ \mathbf{O} & \tilde{\tilde{\mathbf{H}}}_{t-1} \end{pmatrix} = \begin{pmatrix} \quad & \quad \\ \quad & \quad \end{pmatrix}, \quad \psi_{t-1} = \psi_t + \mathrm{tr}(\mathbf{H}_t^{\prime\top} \mathbf{H}_t^{\prime} \mathbf{\Sigma}_\epsilon^{\frac{\top}{2}} \mathbf{\Sigma}_\epsilon^{\frac{1}{2}}).$$

$$\tilde{\mathbf{H}}_{t,a} \in \mathbb{R}^{N,N}, \quad \tilde{\mathbf{H}}_{t,s} \in \mathbb{R}^{N,2N+\rho+1}, \quad \tilde{\tilde{\mathbf{H}}}_{t-1} \in \mathbb{R}^{2N+\rho+1,2N+\rho+1},$$

where $\tilde{\mathbf{H}}$ is the matrix from the preceding step of the Riccati recursion, gained from

$$\tilde{\mathbf{H}}_t = \mathbf{H}_t^{\prime} \begin{pmatrix} \mathbf{B} & \mathbf{A}_{\tau-1} \end{pmatrix},$$

and uses the certainty-equivalence approximation. The multiplier λ can be found explicitly. An additional rotation and (unconstrained) optimization provide the updated loss-to-go,

$$\left\| \begin{pmatrix} \tilde{\tilde{\mathbf{H}}}_{t-1} \\ \mathbf{1}^\top \frac{e_N^\top + \mathbf{1}^\top \tilde{\mathbf{H}}_{t,a}^{-1} \tilde{\mathbf{H}}_{t,s}}{\mathbf{1}^\top \tilde{\mathbf{H}}_{t,a} \mathbf{1}} \end{pmatrix} s_{t-1} \right\|^2 = \left\| \begin{pmatrix} \quad \\ \quad \end{pmatrix} s_{t-1} \right\|^2 = ||\mathbf{H}_{t-1} s_{t-1}||^2, \quad e_N = \begin{pmatrix} 0 \\ \cdot \\ \cdot \\ 1 \end{pmatrix},$$

where $\mathbf{H}_{t-1} \in \mathbb{R}^{3N+\rho+1,3N+\rho+1}$ is again upper triangular.

The second phase realizes the inner loop of the algorithm, which concerns the inequality constraint $a_t \geq 0$, formally stated as in [12], Section 6.2. At the current real time τ, it uses an iterative algorithm that finds the optimal action that satisfies constraints and minimizes

$$\min_{a \in \Omega_{BE}} f(a), \quad f(a) = \frac{1}{2} a^\top \tilde{\mathbf{H}}_a^\top \tilde{\mathbf{H}}_a a - (\tilde{\mathbf{H}}_s s)^\top \tilde{\mathbf{H}}_a a. \tag{4.2}$$

Moreover, the state $s = s_{t-1} \approx$ the current known state $s_{\tau-1}$, is considered during the dynamic programming, as the final found optimal action corresponds to it. With this approximation, the processing illustrated above on the equality constraint is directly applicable, and preserves the quadratic form of the value function (4.1). Without it, the quadratic form is spoiled.

The augmented Lagrangian, keeps the equality condition and corrects for the inequality condition by introducing a regularization parameter ξ

$$\mathcal{L}(a, \bar{\lambda}, \xi) = f(a) + \mathbf{W}^\top \bar{\lambda} + \frac{\xi}{2} ||\mathbf{W} a||^2, \tag{4.3}$$

with gradient

$$g(a, \bar{\lambda}, \xi) = \nabla \mathcal{L}(a, \bar{\lambda}, \xi) = (\tilde{\mathbf{H}}_a^\top \tilde{\mathbf{H}}_a + \xi \mathbf{W}^\top \mathbf{W}) a - (\tilde{\mathbf{H}}_s s)^\top \tilde{\mathbf{H}}_a a + \mathbf{W}^\top \bar{\lambda}, \tag{4.4}$$

where $\bar{\lambda}$ is the multiplier λ from the last step of the Ricatti recursion. The Karush-Kuhn-Tucker conditions, which have to be satisfied to ensure the uniqueness of the feasible solution, are

$$g(a, \bar{\lambda}, \xi) \geq \mathbf{0}, \quad g(a, \bar{\lambda}, \xi)^\top (a - \mathbf{l}) = 0. \tag{4.5}$$

The algorithm solves for the bound constraint in the inner loop and simultaneously updates the Lagrange multiplier $\bar{\lambda}$ for the equality constraints in the outer loop. The multiplier $\bar{\lambda}$ and soft-max[14] of the action vector a^*, computed previously, where just the equality constraint was considered, are taken as the initial value for the algorithm.

The augmented Lagrangian in the square root formalism converts this problem back into the original minimization (4.2).

$$\mathcal{L}_{aug}(a, \bar{\lambda}, \xi) = f(a) = \frac{1}{2} a^\top \tilde{\mathbf{H}}_{aug}^\top \tilde{\mathbf{H}}_{aug} a - \mathbf{b}_{aug} a, \tag{4.6}$$

where

$$\tilde{\mathbf{H}}_{aug} = \begin{pmatrix} \tilde{\mathbf{H}}_a \\ \sqrt{\xi} \mathbf{1}^\top \end{pmatrix}, \quad \mathbf{b}_{aug} = -\tilde{\mathbf{H}}_a \tilde{\mathbf{H}}_s s - \mathbf{1}^\top \bar{\lambda} - \xi \mathbf{1}^\top,$$

with gradient

$$g = \tilde{\mathbf{H}}_{aug}^\top \tilde{\mathbf{H}}_{aug} a - \mathbf{b}_{aug}. \tag{4.7}$$

At each iteration in the inner loop, the action is updated via the gradient projection method

$$a^{(k+1)} = a^{(k)} - w^{(k)} g(a^{(k)}, \bar{\lambda}^{(k)}, \xi^{(k)}), \quad w^{(k)} = -\frac{g^\top h}{h^\top \tilde{\mathbf{H}}_a^\top \tilde{\mathbf{H}}_a h},$$

where

$$h = \begin{cases} 0 & ; a \leq \mathbf{0} \wedge g \geq \mathbf{0} \\ -g & ; \text{else} \end{cases}.$$

In the outer loop, we update the Lagrange multiplier

$$\bar{\lambda}^{(k+1)} = \bar{\lambda}^{(k)} + \xi^{(k)} \mathbf{1}^\top a^{(k)}.$$

If the constraint violation $\mathbf{1}^\top a^{(k+1)} - 1$ is too large, we adjust the regularization parameter

$$\xi^{(k+1)} = \xi^{(k)} \Xi.$$

The author of the cited book [12] recommends $\xi^{(0)} \in [0.5, 1.5]$, $\Xi \in [2, 10]$ as the values for these hyperparameters. At last, we check for convergence.

4.2 One-Step Loss Propagation

We examine the case, where $h = 1$, effectively collapsing the optimization horizon and basing our decision only on the one-step-forward loss. However, the loss-to-go determining matrix $\mathbf{H}$ is accumulated over time instead, [21] page 58, meaning that at each current time τ, set $\mathbf{H}_{\tau+1} := \tilde{\mathbf{H}}_\tau$. Only one step of the Ricatti recursion is performed in the inner loop of the SMALBE algorithm, then proceed with the outer loop, as presented, to obtain the optimal action satisfying the given constraints.

The matrix $\mathbf{H}$ could be treated in the same way as the parameters θ in Section 2.3 and forgetting should be applied. The loss-to-go $\tilde{\mathbf{H}}_\tau$ taken to the next step is found as the convex combination

[14] In the Thompson sampling, it may be the case that the matrix $\tilde{\mathbf{H}}_a$ is ill-conditioned for inversion, then we take the previous action as the initial guess.

of the loss-to-go matrix from the previous step $\mathbf{H}(\hat{\theta}_{\tau-1})$ and the newly found loss-to-go matrix from the current optimization step $\mathbf{H}(\theta_\tau)$,

$$\tilde{\mathbf{H}}_\tau^{-1} = \varphi \mathbf{H}^{-1}(\hat{\theta}_{\tau-1}) + (1-\varphi)\mathbf{H}^{-1}(\hat{\theta}_\tau).$$

The weight φ is found by minimizing the convex function (2.26)

$$\min_{\varphi} D(\varphi || \varphi^0) - \ln \int \prod_i^2 (p(s \mid \mathbf{H}))^{\varphi_i} ds,$$

where $\varphi_1 = \varphi$, $\varphi_2 = 1-\varphi$, $p(s \mid \mathbf{H}) \propto \exp\left[-\frac{1}{2} s^\top (\mathbf{H}\mathbf{H}^\top)^{-1} s\right]$. After evaluation,

$$F(\varphi) = \varphi \ln\left[\frac{\varphi}{\varphi^0}\right] + (1-\varphi)\ln\left[\frac{1-\varphi}{1-\varphi^0}\right] - \frac{1}{2}\ln|\tilde{\mathbf{H}}^{-1}| + \frac{\varphi}{2}\ln|\mathbf{H}(\hat{\theta}_{\tau-1})| + \frac{1-\varphi}{2}\ln|\mathbf{H}(\hat{\theta}_\tau)|$$

The optimal weight φ is again found as $\varphi^* \in \arg\min_\varphi F(\varphi)$ via a golden-section search, minimization algorithm.

Example 4.1.

The comparison between this one-step optimization is compared to the multi-step optimization, Section 4.1 with $h = 10$. The example uses the same data as in Example 2.1, Section 2.3 with 250 time steps. The initial parameters are set to $\alpha_0 = 0.09, \beta_0 = 0.5, \gamma_0 = 0.41, \mu = 0.9$. The optimized prior, Section 2.4 was utilized here, where the first 10 time steps were used to derive it. The matrix of coefficients $\mathbf{P}$ is taken as the point estimate $\hat{\mathbf{P}}$ from its posterior (2.16).

Fig. 4.1, 4.2 show the cumulative returns of our two models compared to Uniform portfolio, see Chapter 5. This is a simple example and the returns are just illustrative. Note that the multi-step optimization is particularly helpful to the Risk-Seeking strategy, which does not include the estimate of the variance $\hat{\boldsymbol{\Sigma}}$ in its objective function, see Chapter 3. The Risk-Averse portfolio is practically unaffected. This gives us a hint that this approach might be useful, as it significantly increases the speed of the optimization.

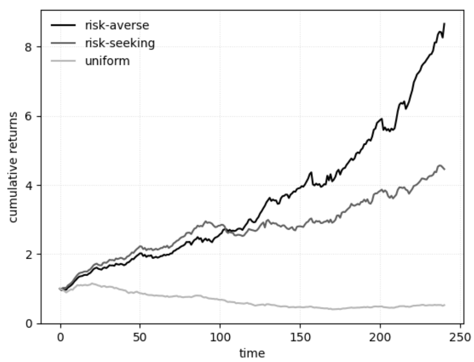


Figure 4.1: Returns with propagating $\mathbf{H}$, one-step optimization.

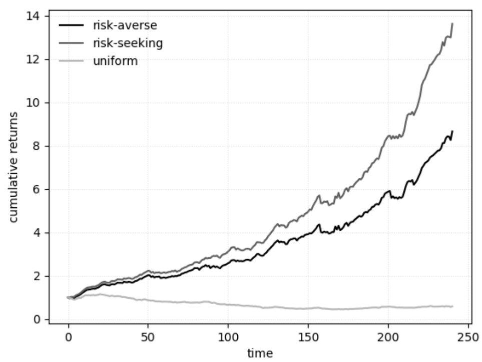


Figure 4.2: Returns with multi-step optimization. At each time step τ, $\mathbf{H}_{\tau+h} = \mathbf{O}$.

5 Experiments

Thorough back-testing and evaluation of any model is a key component of strategy development. Experiments on historical data should be done with a clear purpose behind them. The core aim for us will be to establish the metrics based on which we are going to be evaluating the model's performance, examine how the different hyperparameters might influence these metrics, the overall behavior of our model, and get an intuitive understanding of strengths and weaknesses of our approach.

The hyperparameters left to choose are: all of the parameters for structure estimation Algorithm 2, initial guesses for the forgetting parameters α_0, β_0, Section 2.3, the shrinkage parameter μ, 2.4 and the time horizon h for the optimization, Chapter 4.

We effectively have 2 strategies at hand, the "risk-averse" (RA) strategy, exploiting the FPD in LQR, Section 3 and a "risk-seeking" (RS) strategy, which only cares about maximizing returns. These are enhanced by the new approach to forgetting, Section 2.3 and optimization, Chapter 4. The RS strategy does not take the estimated variance $\hat{\Sigma}$ into consideration, only expected returns $\hat{y}$ and transaction costs. Both strategies are compared to the "Uniform-Start" ($\mathrm{U_S}$) portfolio, which is initiated with uniform allocation and is not actively managed. The second, only theoretical[15] portfolio, is truly "Uniform" (U) in the sense that at all times, the N assets are perfectly diversified, with each asset occupying $1/N$ proportion of the portfolio.

The metrics we are interested in are:

1. maximum draw-down[16] (mdd),
2. variance in returns ($\sigma^2(R)$),
3. average return ($\bar{R}$),
4. Sharpe ratio[17] ($S_R = \frac{\bar{R}}{\sqrt{\sigma^2(R)}}$),
5. total return.

[15] This portfolio is only theoretical, because it does not pay any transaction costs.

[16] The largest percentage loss the portfolio has experienced.

[17] Sharpe ratio is interpreted as the risk-adjusted return, it is calculated as mean return divided by its standard deviation. Some literature subtracts the risk free return from $\bar{R}$ or adjusts the ratio by square-root of the number of steps.

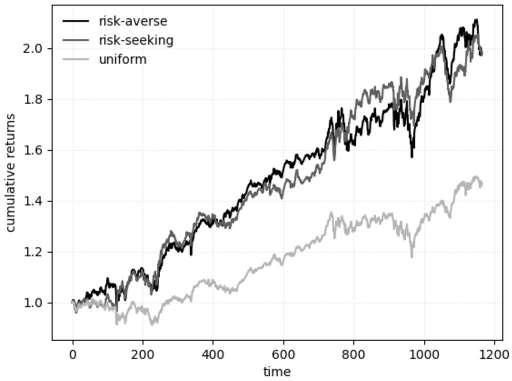
Before we proceed with a large-scale performance test, two key aspects of our strategies need to be addressed. Those are the hyperparameter settings and potential the sampling of matrix of coefficients $\mathbf{P}$ from its posterior pdf (2.16). We show an illustrative example of the typical behavior of these strategies, when Thompson sampling, see Section 3.2 is applied. Recall that the matrix of coefficients $\mathbf{P}$ is sampled and substituted to $\mathbf{A}_{t-1}$, (3.6) at each optimization step. In this example the selected assets to our portfolio were: Standard's&Poors 500 (SPX), Nasdaq 100 (NDX) and gold (GLD), from 2016 to 2021. Fig. 5.1a shows the performance, where $\hat{\mathbf{P}}_t := \mathrm{E}[\mathbf{P} \mid \mathcal{D}^{(t-1)}]$ was taken as its point estimate, for comparison.

batch-size	$\mathrm{p}_{\mathrm{mut}}$	max-iter	alpha	μ	α_0	β_0	γ_0	h
2	0.25	100	0.98	0.1	0.1	0.5	0.4	10

Table 5.1: Hyperparameter setting: Test with sampled $\mathbf{P}$.

$x_t =$	$[y_{t-1}, \ldots, y_{t-5}$	sosc	$1]$

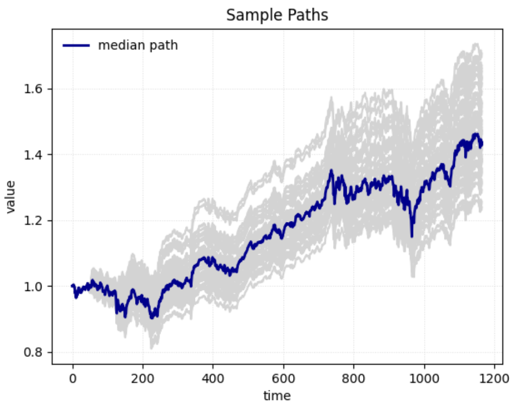
Table 5.2: Regressor $\tilde{x}$: Test with sampled $\mathbf{P}$.



(a) No **P** sampling (Returns)

Metric	RA	RS	U
mdd	-0.126	**-0.110**	-0.130
$\bar{R}$	**6.2e-4**	6.1e-4	3.5e-4
$\sigma^2(R)$	6.9e-5	5.7e-5	**4.3e-4**
S_R	0.075	**0.081**	0.054

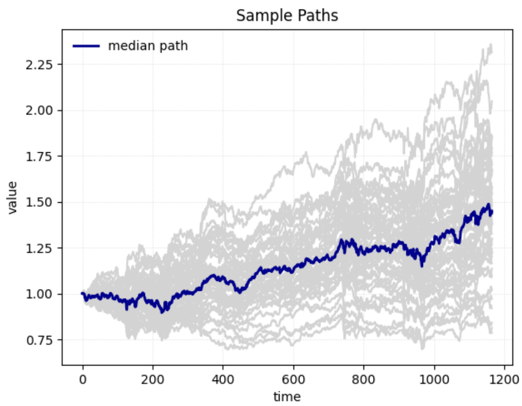
(b) Metrics: No **P** sampling



(c) RA strategy, with **P** sampling

Stat.	mdd	$\bar{R}$	$\sigma^2(R)$	S_R
mean	-0.148	3.50e-4	4.82e-5	0.050
median	-0.151	3.33e-4	4.84e-5	0.040
max	-0.210	4.80e-4	5.24e-5	0.067
min	-0.120	2.10e-4	4.42e-5	0.030

(d) RA Statistics (Sampling **P**)



(e) RS model, with **P** sampling

Stat.	mdd	$\bar{R}$	$\sigma^2(R)$	S_R
mean	-0.210	3.30e-4	8.04e-5	0.037
median	-0.205	3.60e-4	8.03e-5	0.040
max	-0.340	7.60e-4	8.90e-5	0.080
min	-0.110	1.60e-4	7.20e-5	-0.020

(f) RS Statistics (Sampling **P**)

Figure 5.1: Comparative analysis of Risk-Averse and Risk-Seeking strategies under sampling of coefficients matrix.

5.1 200 Portfolios with Custom Regressor

In order to make the asset selection for our tests as unbiased as possible, we took all of the stocks from the S&P500[18] and selected only 20 stocks by the following procedure. First, we filtered out the top 100 performing companies in 2015, of those 100 we have further selected the 20 least pair-wise correlated. Given these 20 stocks, 200 portfolios have been chosen at random and optimized between the years 2023-2025, each portfolio containing $N \in \{8, 9, 10, 11, 12\}$ assets. The selected regressors for testing is

$x_t =$	$[y_{t-1}, \ldots, y_{t-3}$	macd	rolling-var]

Table 5.3: Expertly selected regressor: Test with 200 portfolios.

It is important to restate the assumptions we made in Introduction, Chapter 1. We assume, the investor has low enough amounts of capital such that any order can be filled at the exact market price. Secondly, the investor does not influence the market in any way. These assumptions are quite strong.

At the beginning of each trading day t, the new optimal allocation a_t is made based on the updated parameter estimates. The transaction costs $trc_t = \sum_{i=1}^{N} c_{t,i}|a_{t,i} - a_{t-1,i}|$ are paid immediately at time of this re-balancing. At the end of the trading day, the investor receives their return $R_t = a_t^\top y_t$. The total real return for day t is then $(1 + R_t)(1 - trc_t)$. When the after-market[19] closes, the model's parameters are updated, and new allocation a_{t+1} is calculated to be adopted on the market open the next day.

We ran this simulation twice, once comparing our strategies with the Uniform portfolio, the other time with the Uniform Start portfolio. Hence, we have got the slight differences in metrics.

The hyperparameters α_0, β_0, μ were selected based on a full-factorial design of experiments, [5] performed on data from 2021. The response variable in this task was the Sharpe ratio S_R. The parameter tuning revealed an interesting dynamic between the models and sampling. It suggests that the Risk-Averse strategy performs better when $\mathbf{P}$ is chosen as its points estimate $\mathrm{E}[\mathbf{P} \mid \mathcal{D}^{(t-1)}]$, while Risk-Seeking strategy has better performance under sampling of $\mathbf{P}$ from its posterior, Section 3.2. This aligns with our discussion in Chapter 3, Section 3.2, that Thompson's sampling may be able to substitute for the explicit involvement of the estimated variance $\hat{\mathbf{\Sigma}}$ in the loss function.

The initial choice of $\alpha_0, \beta_0, \gamma_0$ were deemed statistically insignificant. However, μ has an influence on the response S_R. The search suggests that values $\mu \in (0.7, 0.95)$ are reasonable. Moreover, S_R increased slightly and linearly with increasing $\mu \in (0, 1)$.

[18] The S&P500 (Standard and Poor's 500 index) is an index fund diversified across the 500 largest companies by market capitalization in the United States.

[19] Participants on the market may trade derivatives on the underlying assets causing the price to move after the trading hours. Our pipeline uses the adjusted closes for its calculations.

batch-size	p-mut	max-iter	decay	μ	α_0	β_0	γ_0	h
4	0.5	500	0.992	0.9	0.09	0.5	0.41	10

Table 5.4: Large test hyperparameter settings.

The case $h = 1$, explored in Section 4.2, was left aside for future texts.

Tables 5.2b, 5.2a show that our strategy, despite paying transaction costs, behaves similarly to the ideal U portfolio that does not pay these costs. We are truly risk-averse as we improve maximum draw-down and consequently also total return. Compared to the $\mathrm{U_S}$ portfolio, our improvements are even more visible.

	RA	U	RS
mdd	**−0.259**	−0.271	−0.490
total return	**1.269**	1.265	1.195
$\bar{R}$	**5.68e-4**	**5.68e-4**	7.40e-4
$\sigma^2(R)$	1.84e-4	**1.83e-4**	7.30e-4
S_R	**0.042**	**0.042**	0.027

(a) Medians (vs. Uniform benchmark)

	RA	U	RS
mdd	**−0.273**	−0.280	−0.480
total return	1.280	1.262	**1.420**
$\bar{R}$	**5.62e-4**	5.37e-4	8.13e-4
$\sigma^2(R)$	1.88e-4	**1.86e-4**	7.54e-4
S_R	**0.041**	0.039	0.029

(b) Means (vs. Uniform benchmark)

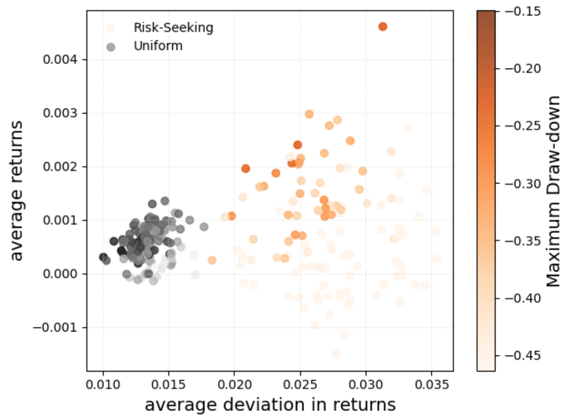
	RA	$\mathrm{U_S}$	RS
mdd	**−0.316**	−0.499	−0.473
total return	**1.180**	0.915	1.060
$\bar{R}$	**5.17e-4**	2.55e-4	5.55e-4
$\sigma^2(R)$	**2.75e-4**	8.49e-4	7.90e-4
S_R	**0.031**	0.009	0.020

(c) Medians (vs. Uniform Start)

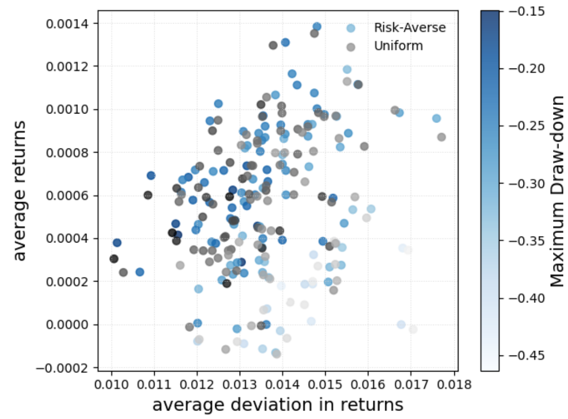
	RA	$\mathrm{U_S}$	RS
mdd	**−0.329**	−0.514	−0.479
total return	1.210	**1.222**	1.260
$\bar{R}$	**4.58e-4**	4.56e-4	5.40e-4
$\sigma^2(R)$	**2.94e-4**	9.45e-4	7.80e-4
S_R	**0.027**	0.015	0.019

(d) Means (vs. Uniform Start)

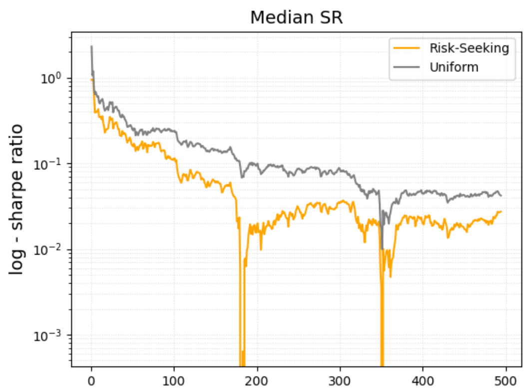
Figure 5.2: Metrics, 200 portfolios



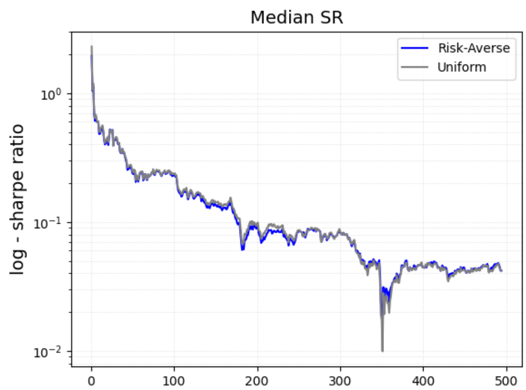
(a) RS vs. Uniform: $\bar{R}$ vs. $\sigma(R)$



(b) RA vs. Uniform: $\bar{R}$ vs. $\sigma(R)$



(c) Sharpe Ratio S_R: RS vs. Uniform



(d) Sharpe Ratio S_R: RA vs. Uniform

Figure 5.3: Performance metrics comparison against the Uniform benchmark.

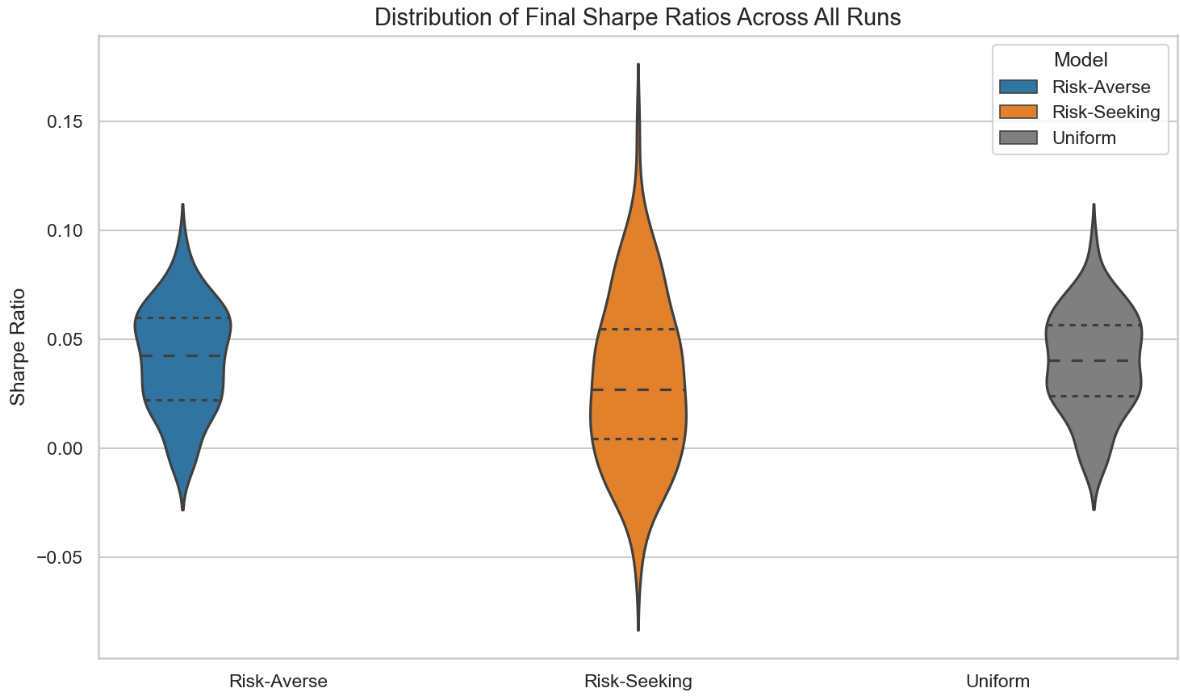
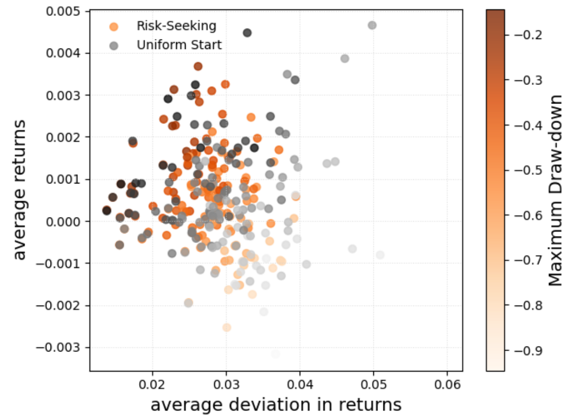
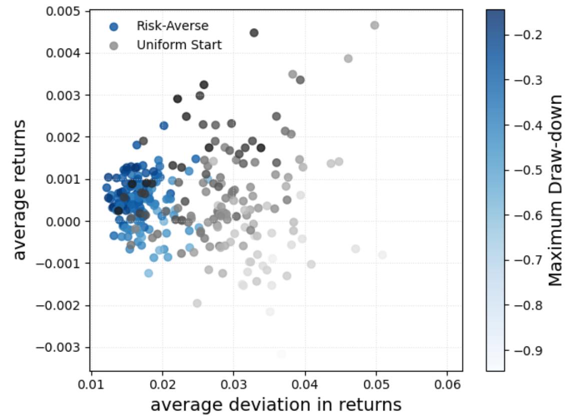


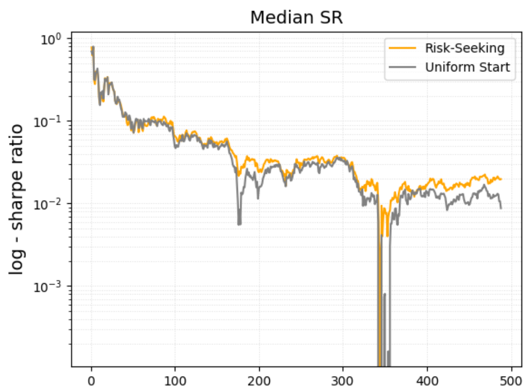
Figure 5.4: Comparison of final Sharpe Ratios.



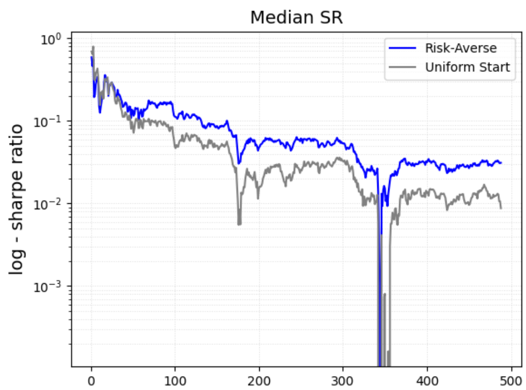
(a) RS vs. U_S: $\bar{R}$ vs. $\sigma(R)$



(b) RA vs. U_S: $\bar{R}$ vs. $\sigma(R)$



(c) Sharpe Ratio S_R: RS vs. U_S



(d) Sharpe Ratio S_R: RA vs. U_S

Figure 5.5: Performance metrics comparison against the Uniform Start (U_S) benchmark.

The important takeaway is that both our strategies outperform the Uniform Start portfolio in median mdd, total return and $\bar{R}$. The Risk Averse model beats both benchmarks in mean and median mdd, and crucially in S_R. It beats or gets very close to the mean and median average returns $\bar{R}$ and variance $\sigma^2(R)$ across all runs produced by the theoretically ideal Uniform portfolio.

5.2 200 Portfolios with Structure Estimation

This section utilizes the proposed Genetic Algorithm 2, Section 2.5. The regressors in this experiment are chosen entirely by the structure estimation, without expert adjustment. See, Appendix *B* for information about what inputs were considered into $\tilde{x}_t$.

We added the conditional Markowitz's portfolio (M), [26] as an additional baseline here. Let $\mathcal{D}^{(t-1)} = (x_{t-L:t-1}, y_{t-L:t-1})$ represent the stacked matrices of historical features and returns over $[t-L, t-1]$, where $L = 40$ is the look-back window and t is the current time. At each time step t, a linear mapping between the historical features and returns is estimated via Ordinary

Least Squares.

$$\hat{\mathbf{P}}_{M;t-1} = (x_{t-L:t-1}^{\top} x_{t-L:t-1})^{-1} x_{t-L:t-1}^{\top} y_{t-L:t-1}.$$

The conditional expected return vector for time t is then the forecast, using the contemporary feature vector x_t:

$$\hat{y}_{M;t} = x_t \hat{\mathbf{P}}_{M;t-1}$$

The conditional covariance matrix for the Markowitz's optimization is derived from the residuals of the rolling regression, capturing the variance unexplained by the predictive features. Let $\mathbf{E}_t$ be the matrix of in-sample residuals

$$\mathbf{E}_t = y_{t-L:t-1} - x_{t-L:t-1} \hat{\mathbf{P}}_{M;t-1}.$$

The sample covariance matrix of these residuals is calculated as $\hat{\mathbf{\Sigma}}_{M;t,raw} = \mathrm{Cov}(\mathbf{E}_t)$. To guarantee that the covariance matrix is strictly positive definite and invertible during optimization, a ridge penalty is applied to the diagonal

$$\hat{\mathbf{\Sigma}}_{M;t} = \hat{\mathbf{\Sigma}}_{M;t,raw} + \delta \mathbf{I}_N$$

where $\delta = 10^{-8}$. Given a specified risk aversion parameter $\omega_M := 2$, the optimal portfolio weights $a_{M;t} \in \mathcal{A}$ are determined by maximizing the conditional risk-adjusted return (the standard Markowitz's objective):

$$a_{M;t} = \arg\max_{a} a^T \hat{y}_{M;t} - \frac{\omega_M}{2} a^T \hat{\mathbf{\Sigma}}_{M;t} a$$

Transaction costs apply.

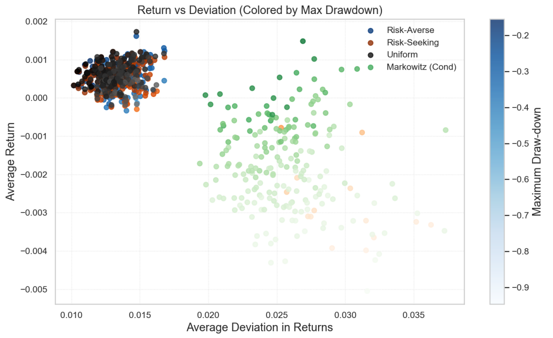
	RA	RS	U	M
mdd	-0.270	-0.311	**-0.268**	-0.705
total return	1.322	1.156	**1.345**	0.421
$\bar{R}$	5.59e-4	2.90e-4	**5.93e-4**	-17.27e-4
$\sigma^2(R)$	1.82e-4	2.28e-4	**1.78e-4**	6.66e-4
S_R	0.042	0.031	**0.045**	-0.067

(a) Metric Means

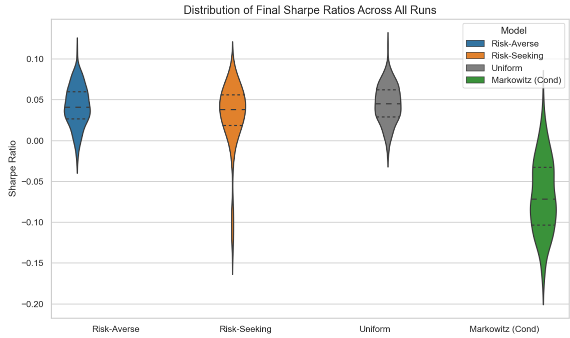
	RA	RS	U	M
mdd	-0.264	-0.273	**-0.262**	-0.739
total return	1.315	1.287	**1.338**	0.406
$\bar{R}$	5.48e-4	5.04e-4	**5.83e-4**	-18.00e-4
$\sigma^2(R)$	1.78e-4	1.78e-4	**1.75e-4**	6.47e-4
S_R	0.041	0.038	**0.046**	-0.071

(b) Metric Medians

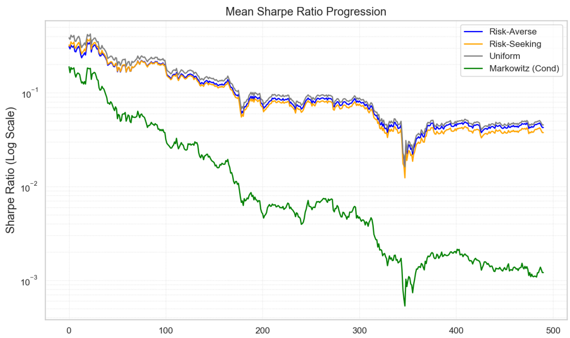
Figure 5.6: Comparison metrics, Test with Structure Estimation.



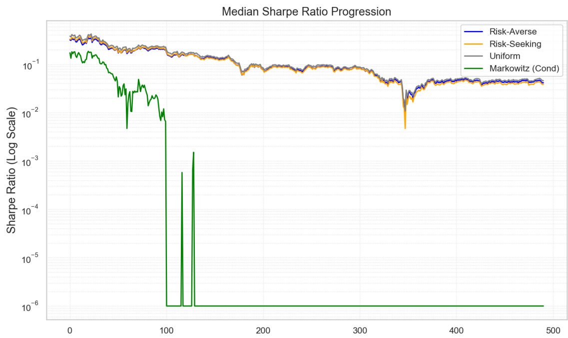
(a) Average Return vs. Deviation



(b) Sharpe Ratio Distribution



(c) Sharpe Ratio Progression (Mean)



(d) Sharpe Ratio Progression (Median)

Figure 5.7: Markowitz optimization analysis and Sharpe ratio progression over time.

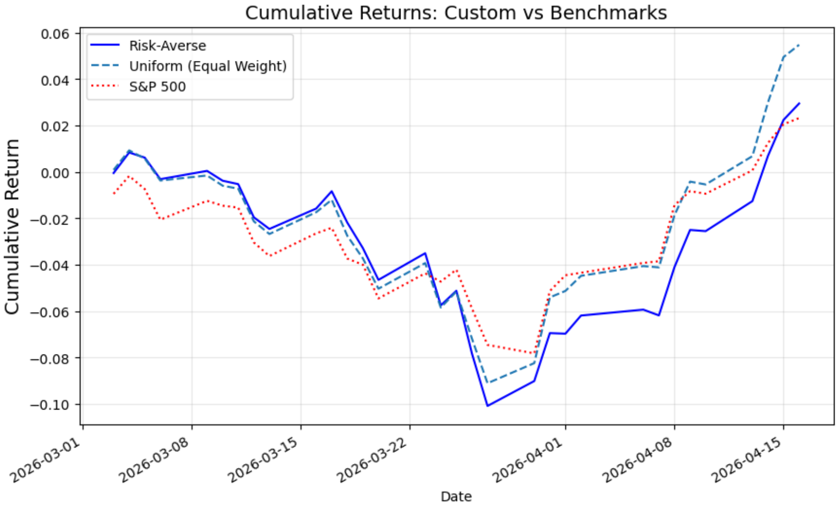
5.3 Live Trading

The live test on real market data ran from March 1st 2026 to April 16th 2026. The selected regressor was

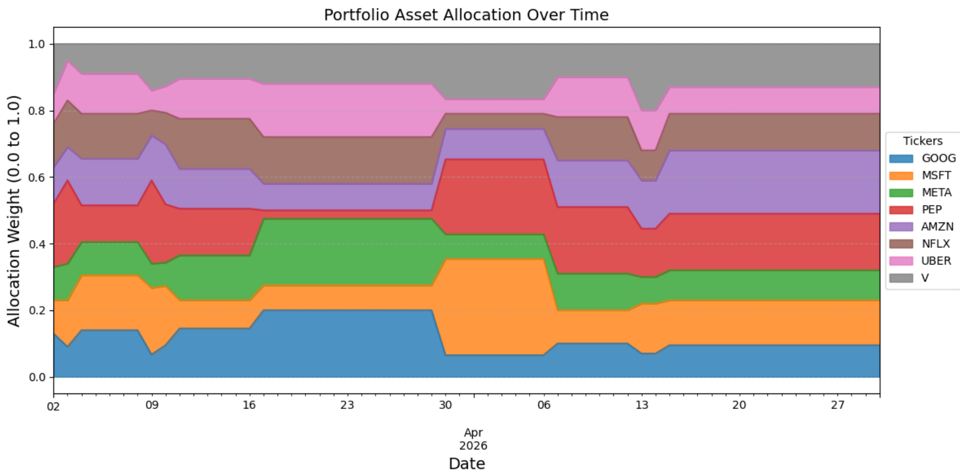
$$x_t = \quad [y_{t-1}, \ldots, y_{t-6} \quad \sin(2\pi t/30)]$$

Table 5.5: x for Live-trading

We have selected a mix of high market cap companies across various sectors and compared the results against the theoretically *optimal* Uniform portfolio and the benchmark ETF (exchange traded fund) S&P500. Fig. 5.8 shows the total returns, Fig. 5.8*a* and the allocation over time, Fig. 5.8*b* of the Risk-Averse strategy. It shows that our strategy is able to adapt to current market conditions by allocating more capital into companies with stable returns, PEP, UBER, V in times of higher uncertainty and re-allocate to 'risk-ON' assets, AMZN, MSFT after recovery. Despite the reactive nature of our algorithm, it was able to slightly outperform the standard benchmark, S&P500 during the test period.



(a) Live-trading returns, comparison with Uniform portfolio and S&P500.



(b) Live-trading, allocation.

Figure 5.8: Live-trading Results

	RA	U	S&P500
mdd	-0.108	-0.099	-0.077
total return	1.023	1.056	1.022
S_R	0.074	0.133	0.069

(a) Live-trading, metrics

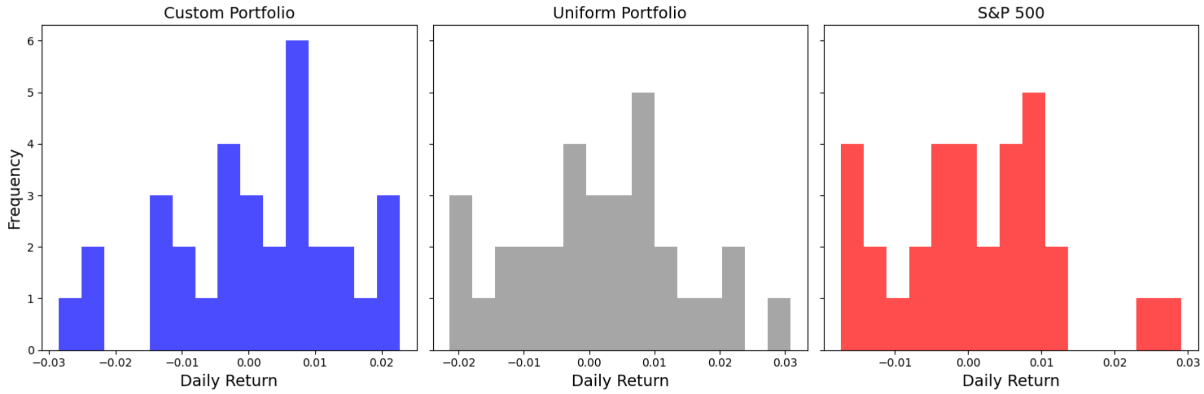


Figure 5.10: Live-trading, Histogram of returns.

Further discussions on the results of experiments are in Conclusions, Chapter 6.

6 Conclusions

This thesis develops a fully probabilistic and adaptive framework for portfolio optimization that integrates Bayesian learning, structure estimation, uncertainty-aware control design, and numerically stable optimization into a unified methodological pipeline, Fig. 1.1.

The core contribution lies in combining:

- multivariate ARX modeling with an optimized conjugate GiW prior, Section 2.1,
- adaptive data-driven forgetting based on minimum relative entropy, Section 2.3,
- a heuristic initiation of the global maximization of structure estimation, Section 2.5,
- fully probabilistic design of risk-aware multi-step decision strategies, Section 3.3,
- square-root based constrained sequential quadratic optimization, Section 4.1.

Each component is derived in a way that preserves consistency, analytical feasibility and methodological completeness, which are the key strengths of our approach. Unlike classical portfolio optimization approaches relying on point estimates of expected returns and covariance, or more recent approaches utilizing neural networks that often obscure risk interpretation, the proposed framework consistently propagates parameter uncertainty into the decision-making process. The fully probabilistic design formulation ensures that risk handling is not heuristic but rather derived directly from a principled and methodological way, incorporating user's aims.

By optimizing forgetting weights, the proposed updating dynamically balances historical information, new observations, and prior regularization. Experimental results demonstrate improved responsiveness in volatile environments, while maintaining stability during stationary periods.

Our connection of fully probabilistic design and linear quadratic regulation, in combination with optimization in square-roots, provides analytical traceability of the solution while preserving numerical stability. This eliminates the need for externally tuned correction mechanisms. At the same time, the number of hyperparameters requiring user intervention is very small, which increases robustness and practical usability.

Empirical evaluations presented in Chapter 5 on real-world financial data suggest that:

- the adaptive Risk-Averse strategy consistently outperforms uniform portfolio benchmark under reasonable conditions with expertly chosen x,

- performance decays when Structure Estimation from Section 2.5 is used,
- classical Markowitz's optimization fails in the long run due to transaction costs,
- uncertainty-aware allocation (the Risk-Averse strategy) improves robustness and successfully manages risks over the Risk-Seeking strategy,
- multi-step optimization with uncertainty inclusion via parameter sampling enhances the strategy optimized under objective function purely focused on maximizing expected returns (the Risk-Seeking strategy).

This framework is modular and extensible. It is not restricted to financial applications and can be adopted in broad adaptive decision-making problems. Although a Gaussian model for returns was adopted for analytical feasibility, the overall pipeline allows for heavy-tailed models.

Future work should focus on extending the modeling layer, while preserving the probabilistic consistency of the pipeline. A natural direction is the incorporation of heavy-tailed return dynamics via adaptively weighted mixtures of Gaussian models or an adaptive regime switching model. In such extension, multiple Gaussian components represent distinct market regimes (e.g., low-volatility, high-volatility, crisis states), with mixture weights updated recursively using Bayesian principles. The function (2.26) could be used, when the predictive distributions (2.9) of the various models are pooled with respective weights and integrated over y. Each component will be estimated using the presented conjugate probability and square-root methods, while the mixture weights would represent regime probabilities. This allows the mixed model to capture heavy tails and structural breaks without abandoning analytical feasibility. The other important area of improvement is the exogenous data considered for $\tilde{x}_t$. So far, only transformations and/or metrics derived from the returns were considered. We want to investigate other data source like quantified text in form of news feed, data from the derivatives markets and expert opinions. Lastly, the extension for larger investors, that may influence the markets. The framework from Section 2.1 already allows for this. What has to be modified is the optimization algorithm from Chapter 4, and a method for optimal re-balancing, with respect to the volume on the market and current order book, needs to be developed.

Empirical tests indicate confidence in the developed methodology.

Appendices

A Lemmas

Lemma A.1. *(Woodbury Identity) Let* $\mathbf{A} \in \mathbb{R}^{n,n}$*,* $\mathbf{B} \in \mathbb{R}^{n,k}$*,* $\mathbf{C} \in \mathbb{R}^{k,k}$*,* $\mathbf{D} \in \mathbb{R}^{k,n}$ *be real matrices and* $\mathbf{A}, \mathbf{C}$ *are non-singular. It is assumed that inversion* $(\mathbf{A} + \mathbf{BCD})^{-1}$ *exists. Then*

$$(\mathbf{A} + \mathbf{BCD})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}(\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1}.$$

Proof: Multiplying the equality by $(\mathbf{A} + \mathbf{BCD})$ from the left side gives

$$\mathbf{I} = \mathbf{I} + \mathbf{BCDA}^{-1} - (\mathbf{B} + \mathbf{BCDA}^{-1}\mathbf{B})(\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1}.$$

Factoring $\mathbf{BC}$ from the first bracket quickly shows that this expression on the right side also reduces to the identity matrix $\mathbf{I}$. □

Lemma A.2. *Let* $\mathbf{A} \in \mathbb{R}^{n,n}$ *be a real non-singular matrix and* $b \in \mathbb{R}^{n}$ *a real vector. Then*

$$\left|\mathbf{A} + bb^\top\right| = |\mathbf{A}|\,(1 + b^\top \mathbf{A}^{-1} b).$$

Proof: We shall show a more general result, from which the proof of this Lemma follows. Let matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}, \mathbf{D}$ be real matrices with appropriate dimensions, matrices $\mathbf{A}$ and $\mathbf{D}$ are non-singular and let

$$\mathbf{E} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}.$$

The matrix $\mathbf{E}$ is LU-decomposed

$$\mathbf{L} = \begin{pmatrix} \mathbf{I} & \mathbf{O} \\ \mathbf{CA}^{-1} & \mathbf{I} \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{O} & \mathbf{D} - \mathbf{CA}^{-1}\mathbf{B} \end{pmatrix},$$

hence $\mathbf{E} = \mathbf{LU}$. Then the trivial property of determinant $|\mathbf{E}| = |\mathbf{L}||\mathbf{U}|$ is applied, which gives the right side of the equality, when we substitute $\mathbf{B} = b$, $\mathbf{C} = -b^\top$, $\mathbf{D} = 1$. To get the left side, we choose

$$\mathbf{L} = \begin{pmatrix} \mathbf{I} & \mathbf{BD}^{-1} \\ \mathbf{O} & \mathbf{I} \end{pmatrix}, \quad \mathbf{U} = \begin{pmatrix} \mathbf{A} - \mathbf{BD}^{-1}\mathbf{C} & \mathbf{O} \\ \mathbf{C} & \mathbf{D} \end{pmatrix}.$$

□

Lemma A.3. *(Linear Combination of Matrices) Let* $\mathbf{A}, \mathbf{B}$ *be matrices of identical dimensions with their respective upper triangular square roots* $\mathbf{A}^{\frac{1}{2}}, \mathbf{B}^{\frac{1}{2}}$ *and let* $a, b \in \mathbb{C}$*. Then the following identity for their linear combination holds*

$$a\mathbf{A} + b\mathbf{B} = \left|\left|\begin{pmatrix} \sqrt{a}\mathbf{A}^{\frac{\top}{2}} & \sqrt{b}\mathbf{B}^{\frac{\top}{2}} \end{pmatrix}\right|\right|^2.$$

Where:

$$\left\|\begin{pmatrix}\sqrt{a}\mathbf{A}^{\frac{\top}{2}} & \sqrt{b}\mathbf{B}^{\frac{\top}{2}}\end{pmatrix}\right\|^2 = \begin{pmatrix}\sqrt{a}\mathbf{A}^{\frac{\top}{2}} & \sqrt{b}\mathbf{B}^{\frac{\top}{2}}\end{pmatrix}\begin{pmatrix}\sqrt{a}\mathbf{A}^{\frac{1}{2}}\\ \sqrt{b}\mathbf{B}^{\frac{1}{2}}\end{pmatrix} = a\mathbf{A}^{\frac{\top}{2}}\mathbf{A}^{\frac{1}{2}} + b\mathbf{B}^{\frac{\top}{2}}\mathbf{B}^{\frac{1}{2}} = a\mathbf{A} + b\mathbf{B}.$$

Lemma A.4. *Let $\mathbf{A} \in \mathbb{R}^{k,k}, \mathbf{B} \in \mathbb{R}^{n,n}$ be real matrices, $\mathbf{A}, \mathbf{B} \geq 0$ and $\mathbf{C}$, $\hat{\mathbf{C}} \in \mathbb{R}^{k,n}$, $n, k \in \mathbb{N}$. Then*

$$\int_{\mathbb{R}^{k,n}} \exp\left[-\frac{1}{2} tr(\mathbf{A}^{-1}(\mathbf{C}-\hat{\mathbf{C}})^T\mathbf{B}(\mathbf{C}-\hat{\mathbf{C}}))\right] d\mathbf{C} = (2\pi)^{\frac{kn}{2}}|\mathbf{A}^{-1}|^{-\frac{n}{2}}|\mathbf{B}|^{-\frac{k}{2}}.$$

Lemma A.5. *Let $\mathbf{A}, \mathbf{D} \in M_+$ be real definite matrices, $M_+ = \{\mathbf{M} \in \mathbb{R}^{k,k} \mid \mathbf{M} > 0\}$, $\Delta \in \mathbb{N}$. Then*

$$\int_{M_+} |\mathbf{A}^{-1}|^{\frac{\Delta}{2}} \exp\left[-\frac{1}{2} tr(\mathbf{A}^{-1}\mathbf{D})\right] d\mathbf{A} = (2^{\Delta+k+1}\pi^{\frac{k-1}{2}})^{\frac{k}{2}}|\mathbf{D}|^{-\frac{\Delta+k+1}{2}} \prod_{j=1}^{k} \Gamma(\tfrac{\Delta+k+2-j}{2}).$$

For proofs of the last two Lemmas, see, for example [4], Section 2.3, 7.2.

Definition A.6. *(Matrix by scalar derivative) Let $\mathbf{A} \in \mathbb{R}^{m,n}$, $\omega \in \mathbb{R}$,*

$$\frac{d}{d\omega}\mathbf{A}(\omega) = \begin{pmatrix} \frac{da_{11}}{d\omega} & \dots & \frac{da_{1n}}{d\omega} \\ \vdots & & \vdots \\ \frac{da_{m1}}{d\omega} & \dots & \frac{da_{mn}}{d\omega} \end{pmatrix}.$$

Lemma A.7. *(Jacobi's Determinant Derivative) Let $\mathbf{A} \in \mathbb{R}^{n,n}$ be an invertible matrix, $n \in \mathbb{N}$ and let $\omega \in \mathbb{R}$ be a scalar variable. Then*

$$\frac{d}{d\omega}|\mathbf{A}(\omega)| = |\mathbf{A}(\omega)| tr\left(\mathbf{A}^{-1}(\omega)\frac{d}{d\omega}\mathbf{A}(\omega)\right)$$

For proof, see for example, [8].

Lemma A.8. *(Matrix Inverse Derivative Lemma) Let $\mathbf{A} \in \mathbb{R}^{n,n}$ be an invertible real matrix, $n \in \mathbb{N}$ and let $\omega \in \mathbb{R}$ be a scalar variable. Suppose that $\frac{d}{d\omega}\mathbf{A}(\omega)$ exists. Then $\frac{d}{d\omega}\mathbf{A}^{-1}(\omega)$ exists and the following equality holds*

$$\frac{d}{d\omega}\mathbf{A}^{-1}(\omega) = -\mathbf{A}^{-1}(\omega)\frac{d}{d\omega}\mathbf{A}(\omega)\mathbf{A}^{-1}(\omega).$$

Proof of identity:

$$\mathbf{O} = \frac{d}{d\omega}\mathbf{I} = \frac{d}{d\omega}(\mathbf{A}(\omega)\mathbf{A}^{-1}(\omega)) = \frac{d}{d\omega}\mathbf{A}(\omega)\mathbf{A}^{-1}(\omega) + \mathbf{A}(\omega)\frac{d}{d\omega}\mathbf{A}^{-1}(\omega)$$

$$\Rightarrow \frac{d}{d\omega}\mathbf{A}^{-1}(\omega) = -\mathbf{A}^{-1}(\omega)\frac{d}{d\omega}\mathbf{A}(\omega)\mathbf{A}^{-1}(\omega)$$

□

The proof can also be explicitly shown from the definition of derivative. What remains to be show is the existence of the derivative of the inverse, which can be found in the many books on matrix calculus like, [25] or [8].

Lemma A.9. *(Completion to Squares) Let $a, b, c \in \mathbb{R}^n$, $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{n,n}$, $\mathbf{A} > 0, \mathbf{B} \geq 0$ or $\mathbf{A} \geq 0, \mathbf{B} > 0$, $\mathbf{A}^\top = \mathbf{A}$, $n \in \mathbb{N}$. Let $\mathbf{C} := \mathbf{A} + \mathbf{B}$ and let $\mathbf{C}, \mathbf{A}$ commute. Then*

1. $(a-b)^\top \mathbf{A}(a-b) - a^\top c = \left(a - b - \frac{1}{2}\mathbf{A}^{-1}c\right)^\top \mathbf{A} \left(a - b - \frac{1}{2}\mathbf{A}^{-1}c\right) - b^\top c - \frac{1}{4}c^\top \mathbf{A}^{-1}c$

2. $(a-b)^\top \mathbf{A}(a-b) + a^\top \mathbf{B}a = (a - \mathbf{C}^{-1}\mathbf{A}b)^\top \mathbf{C}(a - \mathbf{C}^{-1}\mathbf{A}b) + b^\top(\mathbf{A} - \mathbf{A}\mathbf{C}^{-1}\mathbf{A})b$

Proof follows simply from writing out both sides of the equations. We imposed $\mathbf{A}^\top = \mathbf{A}$ because that is the case in the text, where the lemma is used. From $\mathbf{A}, \mathbf{B} > 0$ follows $\mathbf{C} > 0$, hence invertible.

B Data

List of selected stocks for extensive test in Chapter 5: MRNA, COP, MOH, AXON, WST, EQT, AZO, DXCM, DELL, EXR, ENPH, BG, CRWD, EPAM, OXY, TGT, TSLA, CF, ANET, TSCO. Daily data of returns and volumes[20], from Jan 1st. 2015 - Jan 1st. 2025, were sourced from Yahoofinance via Python. $\tilde{x}$ contains lagged returns of up to past 10 trading days.

Used financial indicators[21] are the moving average exponential (mae), rolling draw-down, rolling variance in the returns, macd, stochastic oscillator (sosc) and custom relative strength index (rsi).

$$\mathrm{mae}_t(p) = \alpha y_t + (1-\alpha)\mathrm{mae}_{t-1}, \quad \alpha = \frac{2}{p+1}$$

$$\mathrm{macd}_t = \mathrm{macd}_t(p_1) - \mathrm{mae}_t(p_2), \quad p_1 > p_2$$

$$\mathrm{sosc}_t(p) = \frac{|y_t - \mathrm{low}(p)|}{|\mathrm{high}(p) - \mathrm{low}(p)|},$$

where $\mathrm{low}(p), \mathrm{high}(p)$ are the lowest and highest, respectively, among the p-last values.

$$\begin{aligned}\mathrm{rsi}_t(p) = &\left(1 + \exp\left[0.4 * 100\left(1 - \frac{1}{1 + \frac{\mathrm{avgG}(p)}{\mathrm{avgL}(p)}}\right) - 30)\right]\right)^{-1} \\ &+ \left(1 + \exp\left[0.4 * 100\left(1 - \frac{1}{1 + \frac{\mathrm{avgG}(p)}{\mathrm{avgL}(p)}}\right) - 70)\right]\right)^{-1} - 1,\end{aligned} \tag{B.1}$$

where $\mathrm{avgG}, \mathrm{avgL}$ are the average gains and losses, respectively, for the period of last p observations. We took values $p_1 = 10$, $p_2 = 5$.

Other metrics considered into the regressor x_t are the rolling variance of the returns, rolling draw-down of the past p returns of individual assets, sine functions with periods $10, 30, 90$ and intercept 1.

Fig. B.1 shows the correlation matrix of the selected tickers.

[20] We ended up not using volume data due to unreliability of the data and zero inflation. Meaning volumes were not consistent with data from the broker and contained many zero entries.

[21] Indicators are transformation of financial data like returns and volumes that may or may not carry predictive power.

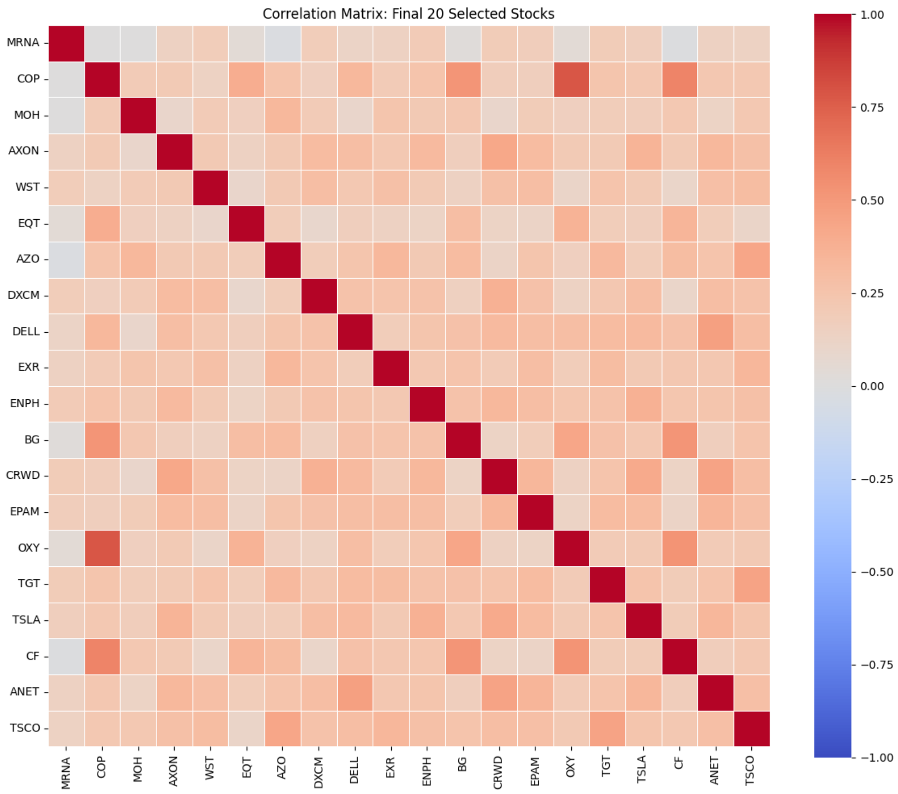


Figure B.1: Correlation matrix: selected tickers.

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